

EMC Test Report

Product: Portable air conditioner

Model: Refer to model list on section 1.4.1

Applicant: Zhejiang Oulun Electric Co., Ltd.

Building 1-8, No. 22 Hengyi Street, Tangqi Town
Linping District
311100 Hangzhou, Zhejiang
PEOPLE'S REPUBLIC OF CHINA



**Add value.
Inspire trust.**

In accordance with EN IEC 55014-1,
EN IEC 55014-2, EN IEC 61000-3-2 and
EN 61000-3-3

COMMERCIAL-IN-CONFIDENCE

Issue Date: Feb. 26, 2026

Report Number: 878882357001-03

RESPONSIBLE FOR	NAME	SIGNATURE	DATE
Approved By	Hua ZHANG	 	Feb. 26, 2026
Prepared By	Saisai YANG	 	Feb. 26, 2026

Signatures in this approval box have checked this document in line with the requirements of TÜV SÜD Product Service control rules.

EXECUTIVE SUMMARY

Three sample of this product were tested and found to be in compliance with EN IEC 55014-1:2021, EN IEC 55014-2:2021, EN IEC 61000-3-2:2019/A2:2024 and EN 61000-3-3:2013/A2:2021.

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1 Report Summary

1.1 Report Modification Record

Alterations and additions to this report will be issued to the holders of each copy in the form of a complete document.

Issue	Description of Change	Date of Issue
1	First Issue	Feb. 26, 2026

1.2 Introduction

The information contained in this report is intended to show verification of the EMC Qualification Approval Testing of the requirements of the standards for the tests listed in Section 1.3.

Applicant	Zhejiang Oulun Electric Co., Ltd.
Address	Building 1-8, No. 22 Hengyi Street, Tangqi Town Linping District 311100 Hangzhou, Zhejiang PEOPLE'S REPUBLIC OF CHINA
Manufacturer	same as above
Address	same as above
Factory	same as above
Address	same as above
Model Number(s)	Refer to model list on section 1.4.1
Rated input voltage	220-240V~
Rated input frequency	50Hz
Rated input power	OL-A016xy15N2, OL-A016xy15N2/S: Cooling: 560W; OL-A016xy20RN2, OL-A016xy20RN2/S: Cooling: 780W, Heating: 685W; OL-A016xy20N2, DCAC07, OL-A016xy20N2/S: Cooling: 780W; OL-A016xy23RN2, OL-A016xy23RN2/S: Cooling: 870W, Heating: 720W; OL-A016xy23N2, OL-A016xy23N2/S: Cooling: 870W; OL-A016xy26RN2, OL-A016xy26RN2/S: Cooling: 1005W, Heating: 835W; OL-A016xy26N2, DCAC09, OL-A016xy26N2/S: Cooling: 1005W
Sample Number(s)	NIB-987082-1~3
Number of Samples Tested	3
Test Specification	EN IEC 55014-1:2021, EN IEC 55014-2:2021, EN IEC 61000-3-2:2019/A2:2024 and EN 61000-3-3:2013/A2:2021
Date of Receipt of EUT	12/15/2025
Start of Test	12/17/2025



Finish of Test	02/02/2026
Name of Engineer(s)	Saisai YANG

1.3 Brief Summary of Results

The sample's mentioned in this report is/are submitted/ supplied/ manufactured by client. The laboratory therefore assumes no responsibility for accuracy of information on the brand name, model number, origin of manufacture, consignment or any information supplied.

A brief summary of the tests carried out in accordance with EN IEC 55014-1, EN IEC 55014-2, EN IEC 61000-3-2 and EN 61000-3-3 is shown below.

Section	Specification	Clause	Test Description	Result	Comments/Base Standard
OL-A016PP26RN2, OL-A016JA26RN2: Power on, all function, high & low fan speed, min & mid & max temperature, timer on; OL-A016JC26RN2/S: Power on, all function, high & low fan speed, min & mid & max temperature, timer on, swing					
2.1	EN IEC 55014-1:2021	4.3.3.6	Conducted Disturbance	Pass (Minimum limit margin: >6dB)	
2.2	EN IEC 55014-1:2021	4.4	Discontinuous Disturbance	Pass	
2.3	EN IEC 55014-1:2021	4.3.4.5	Radiated Disturbance	Pass (Minimum limit margin: >6dB)	
2.4	EN IEC 61000-3-2:2019/A2:2024	7	Harmonic current emission	Pass	
OL-A016PP26RN2, OL-A016JA26RN2: Power on, all function, high & low fan speed, mid temperature, timer on; OL-A016JC26RN2/S: Power on, all function, high & low fan speed, mid temperature, timer on, swing					
2.6	EN IEC 55014-2:2021	5.1	Electrostatic discharge immunity test	Pass	IEC 61000-4-2
2.7	EN IEC 55014-2:2021	5.2	Electrical fast transient /burst immunity test	Pass	IEC 61000-4-4
2.8	EN IEC 55014-2:2021	5.3	Immunity to conducted disturbances, induced by radio-frequency fields	Pass	IEC 61000-4-6
2.9	EN IEC 55014-2:2021	5.5	Radiated, radio-frequency, electromagnetic field immunity test	Pass	IEC 61000-4-3
2.10	EN IEC 55014-2:2021	5.6	Surge immunity test	Pass	IEC 61000-4-5
2.11	EN IEC 55014-2:2021	5.7	Voltage dips, short interruptions and voltage variations immunity tests	Pass	IEC 61000-4-11
AC Powered Power on, switch on/off					
2.5	EN 61000-3-3:2013/A2:2021	5	Flicker	Pass	

1.4 Product Information

1.4.1 Technical Description

NOTICE: This report is a SUPPLEMENT OF Report 878882357001-00, 878882357001-01 and 878882357001-02. So this report is not valid without the report of 878882357001-00, 878882357001-01 and 878882357001-02.

According to client's request and declaration:

1. Seven new series model, a new stepping motor, two new display board and two remote control which highlighted in model list were added on the E8A attestation and report.

2. All models have similar electrical schematic diagram except compressor, with or without stepping motor, fan motor, display board and remote control

Detailed information is shown as below:

Model list:

Mode	Compressor	Stepping motor	Fan motor	Display board	Remote control
OL-A016xy15N2	DSK86V01UZZ3	N/A	YSK95-55-4B, YYS55-4	OL-A010-E, OL-A010P-D, OL-A016G-D, OL-A016E-D, OL-A010Q-D, OL-A010T-D, OL-A016N-D, OL-A016W-DISP, OL-A010S-D, OL-A016M-DISP	14J-C, OL-YRQ-002, OL-YRQ-004
OL-A016xy15N2/S		24BYJ48			
OL-A016xy20RN2	N/A				
OL-A016xy20RN2/S	24BYJ48				
OL-A016xy20N2, DCAC07	N/A				
OL-A016xy20N2/S	24BYJ48				
OL-A016xy23RN2	N/A				
OL-A016xy23RN2/S	24BYJ48				
OL-A016xy23N2	N/A				
OL-A016xy23N2/S	24BYJ48				
OL-A016xy26RN2	DSM155V01UCZC, 44P193I, QZ156S301	N/A	YSL55-4K, YYS65-4		
OL-A016xy26RN2/S		24BYJ48			
OL-A016xy26N2, DCAC09		N/A			
OL-A016xy26N2/S		24BYJ48			
(x=A-Z, y=A-Z, represent different appearances)					

So model

OL-A016PP26RN2 (Compressor: QZ156S301, Fan motor: YSL55-4K, Display board: OL-A010S-D, Remote control: OL-YRQ-002),

OL-A016JA26RN2 (Compressor: QZ156S301, Fan motor: YSL55-4K, Display board: OL-A016M-DISP, Remote control: OL-YRQ-002),

OL-A016JC26RN2/S (Compressor: DSM155V01UCZC, Fan motor: YSL55-4K, Display board: OL-A016M-DISP, Remote control: OL-YRQ-004)

were chosen to perform all the tests.

Tests are performed under power on, all function (all function include cooling, heating, dehumidifier, sleep), high & low fan speed, min & mid & max temperature, timer on, swing mode, the worst test results are recorded.

1.4.2 EUT Port/Cable Identification

Port	Max Cable Length specified	Usage	Type	Screened
AC Powered Power on, all function, high & low fan speed, min & mid & max temperature, timer on, swing				
AC mains port	1.9 m	Power cord	3 core	No

1.4.3 Test Configuration

Configuration	Description
AC Powered	220-240V~, 50Hz

1.4.4 Modes of Operation

Mode	Description
Power on, timer on, all function, high & low fan speed, min & mid & max temperature, timer on, swing	The EUT was power on, all function, high & low fan speed, min & mid & max temperature, timer on, swing.
switch on/off	The EUT was switch on/off.

1.4.5 Monitoring of Performance

The EUT works normally.

1.4.6 Performance Criteria

Performance criterion A: The apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonable expect from the apparatus if used as intended.

Performance criterion B: The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. During the test, degradation of performance is allowed, however no change of actual operating state or stored data is allowed to persist after test. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and from what the user may reasonable expect from the apparatus if used as intended.

Performance criterion C: Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or by any operation specified in the instruction for use.

1.5 Deviations from the Standard

No deviations from the applicable test standard were made during testing.

1.6 Test Location

The following tests were performed at TÜV SÜD Certification and Testing (China) Co., Ltd. Ningbo branch

Address:

Building 2, Essence Adream of Space II, No.350,
Jinghua Road, National Hi-Tech Industrial Development Zone, Ningbo, Zhejiang, China

Test Name	Name of Engineer(s)
Conducted Disturbance	Peiqiang WANG
Discontinuous Disturbance	Peiqiang WANG
Radiated Emissions	Peiqiang WANG
Harmonic current emission	Peiqiang WANG
Flicker	Peiqiang WANG
Electrostatic discharge immunity test	Peiqiang WANG
Electrical fast transient /burst immunity test	Peiqiang WANG
Immunity to conducted disturbances, induced by radio-frequency fields	Peiqiang WANG
Radiated, radio-frequency, electromagnetic field immunity test	Peiqiang WANG
Surge immunity test	Peiqiang WANG
Voltage dips, short interruptions and voltage variations immunity tests	Peiqiang WANG

2 Test Details

2.1 Conducted Disturbance

2.1.1 Specification Reference

EN IEC 55014-1:2021, Clause 4.3.3.6

2.1.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.1.3 Date of Test

12/19/2025, 12/22/2025, 01/28/2026

2.1.4 Test Method

Table-top EUT shall be placed:

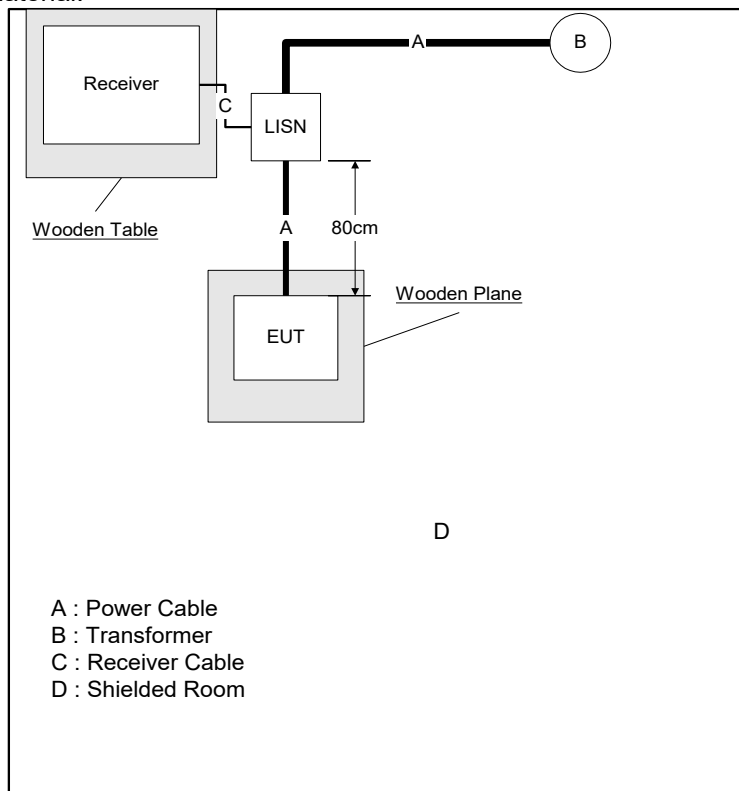
- at a distance of $(0,4 \pm 0,05)$ m from a RGP of at least $2 \text{ m} \times 2 \text{ m}$ in size;
- at a distance of 0,8 m from the AMN, and
- shall be kept at least 0,8 m from any other earthed conducting surface.

The RGP shall be either horizontal or vertical.

Floor standing EUT shall be:

- placed at a height of $(0,12 \pm 0,04)$ m above a horizontal RGP of at least $2 \text{ m} \times 2 \text{ m}$ in size,
- placed at a distance of 0,8 m from the AMN,
- kept at least 0,8 m from any other earthed conducting surface, and
- placed in a manner to ensure that the RGP extends at least 0,5 m beyond the EUT boundaries

Parts supporting the EUT and its parts at the required height shall be made of non-conductive material.



2.1.5 Environmental Conditions

Ambient Temperature 10-35 °C
 Relative Humidity 30-60 %
 Atmospheric Pressure 1010-1040 mbar

2.1.6 Specification Limits

Household appliances and equipment causing similar disturbances and regulating controls incorporating semiconductor devices						
Frequency range	Disturbance voltage at main terminals		Disturbance voltage at associated ports		Disturbance current at associated ports	
MHz	Quasi-peak dB μ V	Average dB μ V	Quasi-peak dB μ V	Average dB μ V	Quasi-peak dB μ A	Average dB μ A
0.15 to 0.5	66 to 56	59 to 46	80	70	40 to 30	30 to 20
0.5 to 5	56	46	74	64	30	20
5 to 30	60	50	74	64		

2.1.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, cooling, high fan speed, min temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, cooling, high fan speed, min temperature, timer on, swing

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

150k-30MHz Conducted Emission Test

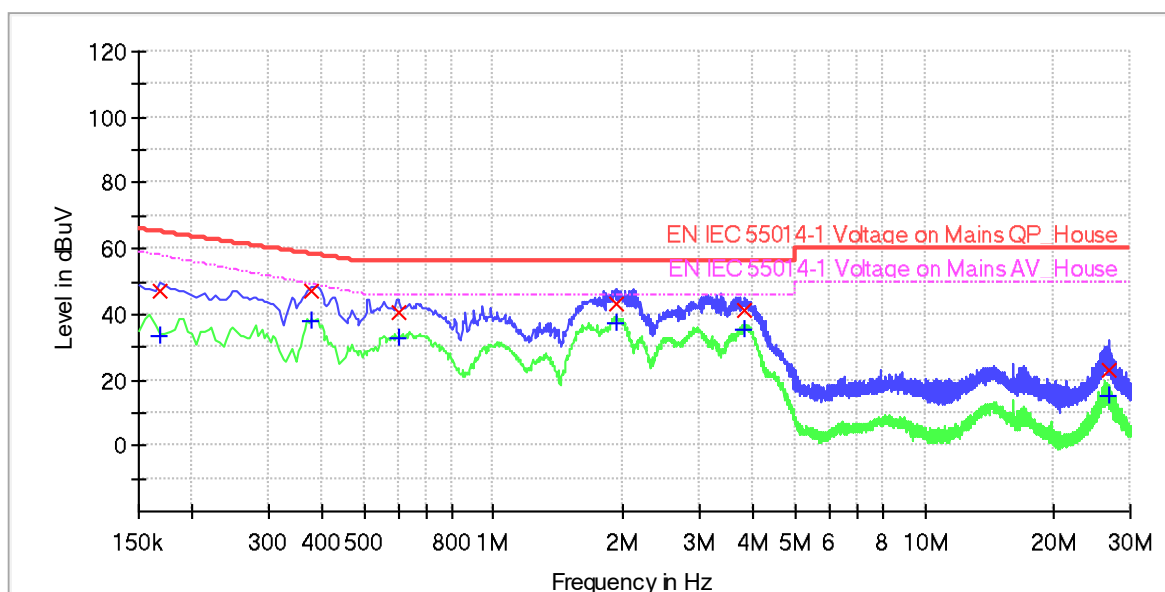
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016PP26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, T29.3°C, H50.3%, P102.2kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: L
 Sample No.: NIB-987082-1

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.168000	46.8	33.6	1000.0	9.000	L1	9.7	18.219	65.059	24.196	57.776
0.379500	47.0	38.0	1000.0	9.000	L1	9.7	11.270	58.290	10.977	48.977
0.600000	40.4	32.8	1000.0	9.000	L1	9.7	15.570	56.000	13.220	46.000
1.918500	43.2	37.5	1000.0	9.000	L1	9.7	12.790	56.000	8.540	46.000
3.835500	41.2	35.5	1000.0	9.000	L1	9.7	14.760	56.000	10.480	46.000
26.628000	22.8	15.2	1000.0	9.000	L1	9.9	37.230	60.000	34.800	50.000

150k-30MHz Conducted Emission Test

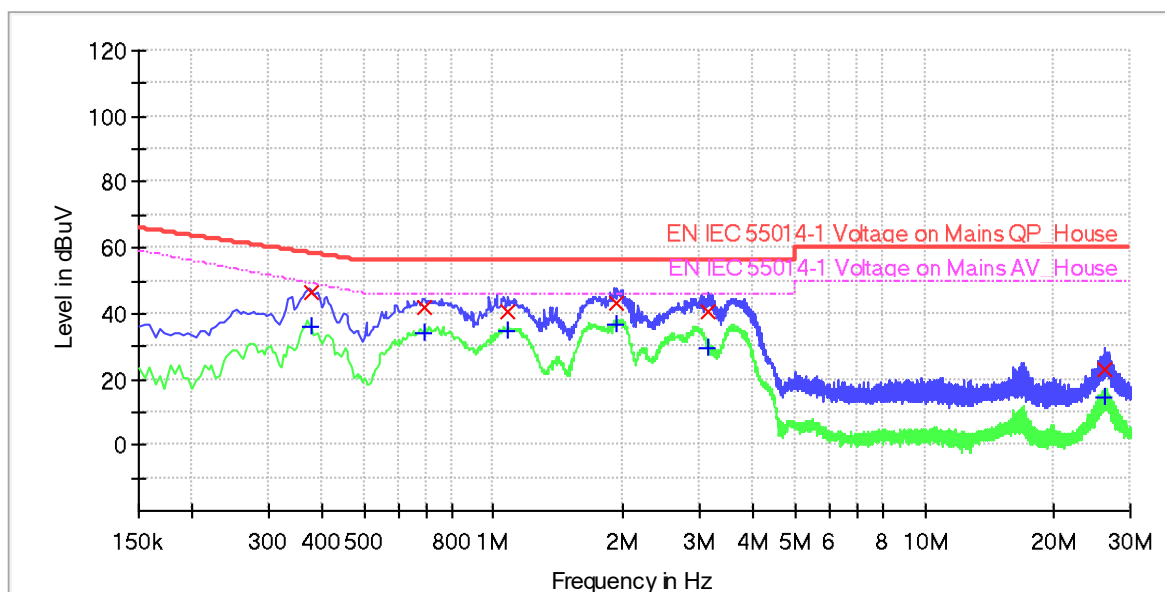
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016PP26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, T29.3°C, H50.3%, P102.2kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: N
 Sample No.: NIB-987082-1

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.379500	46.1	36.1	1000.0	9.000	N	9.7	12.160	58.290	12.837	48.977
0.690000	41.6	34.2	1000.0	9.000	N	9.7	14.440	56.000	11.830	46.000
1.081500	40.7	35.0	1000.0	9.000	N	9.7	15.270	56.000	11.010	46.000
1.936500	43.4	36.9	1000.0	9.000	N	9.7	12.610	56.000	9.110	46.000
3.160500	40.5	29.3	1000.0	9.000	N	9.7	15.470	56.000	16.750	46.000
26.245500	22.7	14.6	1000.0	9.000	N	9.9	37.320	60.000	35.410	50.000

150k-30MHz Conducted Emission Test

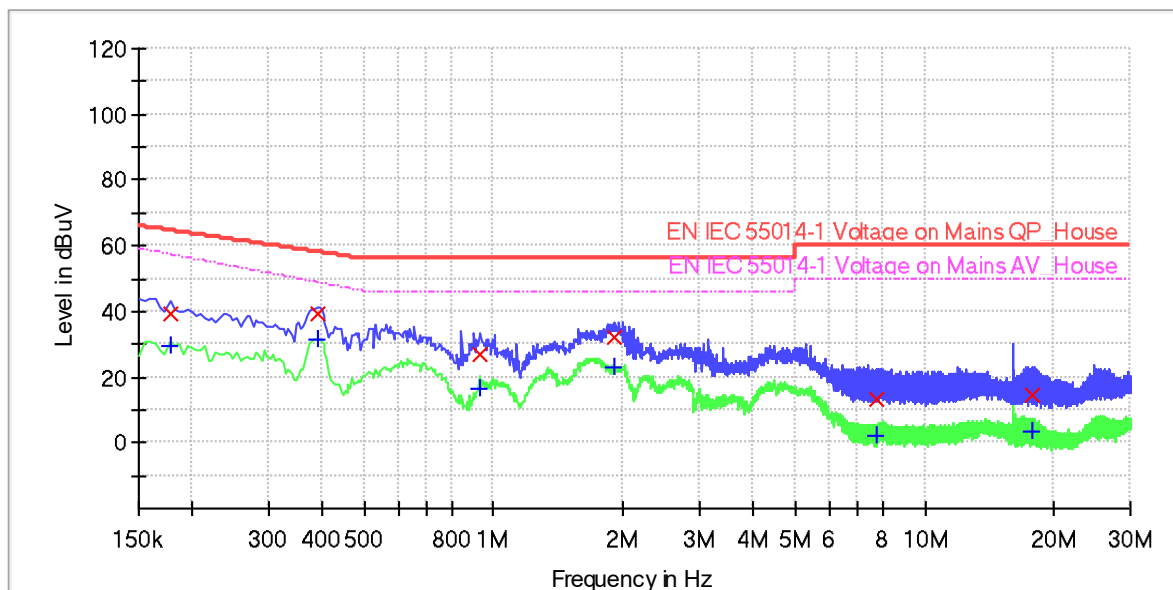
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JA26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, T29.5°C, H50.8%, P102.2kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: L
 Sample No.: NIB-987082-2

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.177000	39.0	29.6	1000.0	9.000	L1	9.7	25.635	64.625	27.623	57.213
0.388500	39.3	31.6	1000.0	9.000	L1	9.7	18.826	58.096	17.174	48.724
0.933000	27.0	16.5	1000.0	9.000	L1	9.7	28.980	56.000	29.470	46.000
1.914000	32.0	22.7	1000.0	9.000	L1	9.7	23.970	56.000	23.320	46.000
7.746000	13.0	2.2	1000.0	9.000	L1	9.7	46.970	60.000	47.760	50.000
17.668500	14.5	3.2	1000.0	9.000	L1	9.8	45.540	60.000	46.810	50.000

150k-30MHz Conducted Emission Test

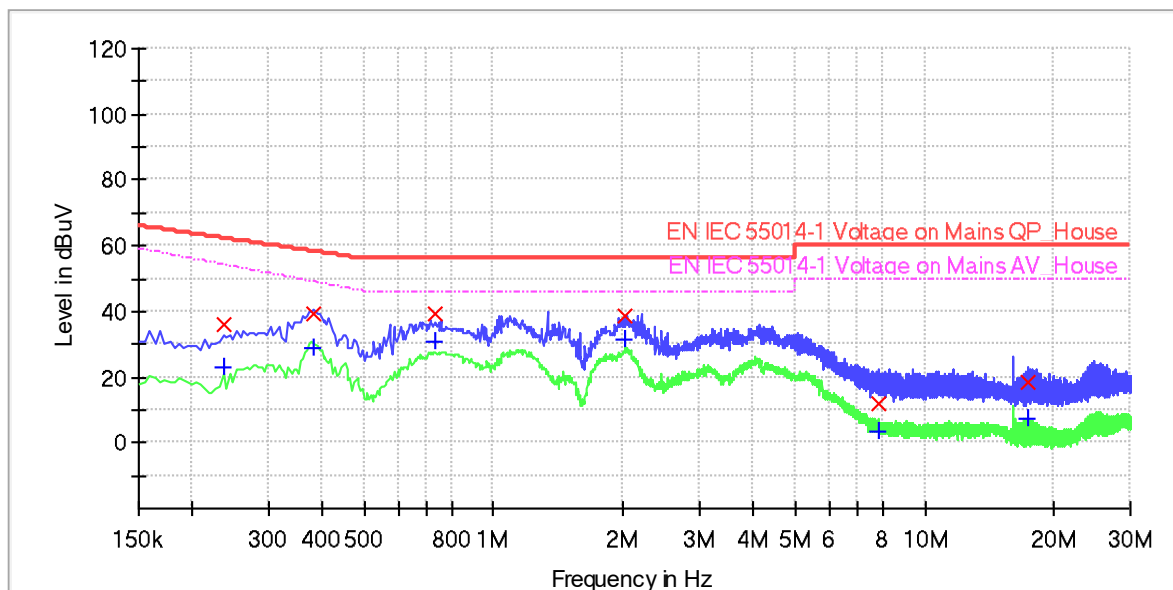
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JA26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, T29.5°C, H50.8%, P102.2kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: N
 Sample No.: NIB-987082-2

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.235500	35.8	23.1	1000.0	9.000	N	9.7	26.463	62.253	30.989	54.129
0.384000	39.5	29.1	1000.0	9.000	N	9.7	18.662	58.192	19.710	48.850
0.735000	39.0	30.7	1000.0	9.000	N	9.7	17.050	56.000	15.260	46.000
2.017500	38.6	31.8	1000.0	9.000	N	9.7	17.430	56.000	14.250	46.000
7.804500	11.8	3.5	1000.0	9.000	N	9.7	48.250	60.000	46.520	50.000
17.295000	18.2	7.5	1000.0	9.000	N	9.8	41.780	60.000	42.460	50.000

150k-30MHz Conducted Emission Test

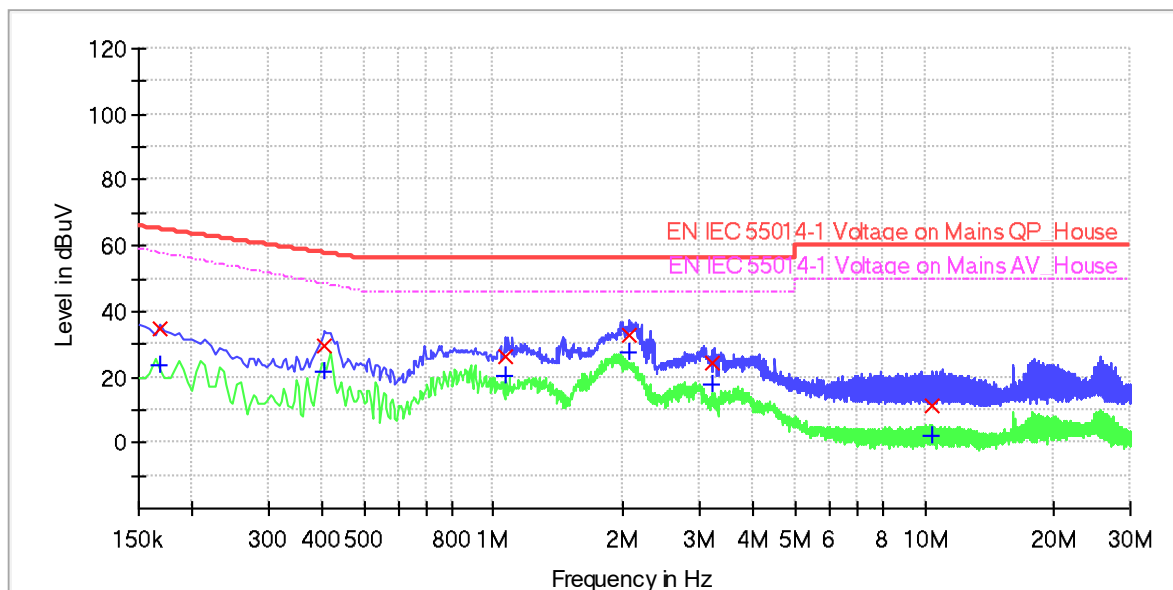
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JC26RN2/S
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, swing, T29.5°C, H51.2%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: L
 Sample No.: NIB-987082-3

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.168000	34.9	23.9	1000.0	9.000	L1	9.7	30.199	65.059	33.886	57.776
0.406500	29.6	22.0	1000.0	9.000	L1	9.7	28.080	57.720	26.285	48.235
1.063500	26.5	20.7	1000.0	9.000	L1	9.7	29.550	56.000	25.320	46.000
2.058000	33.0	27.9	1000.0	9.000	L1	9.7	23.020	56.000	18.140	46.000
3.210000	24.5	17.5	1000.0	9.000	L1	9.7	31.520	56.000	28.550	46.000
10.446000	11.3	1.9	1000.0	9.000	L1	9.8	48.710	60.000	48.090	50.000

150k-30MHz Conducted Emission Test

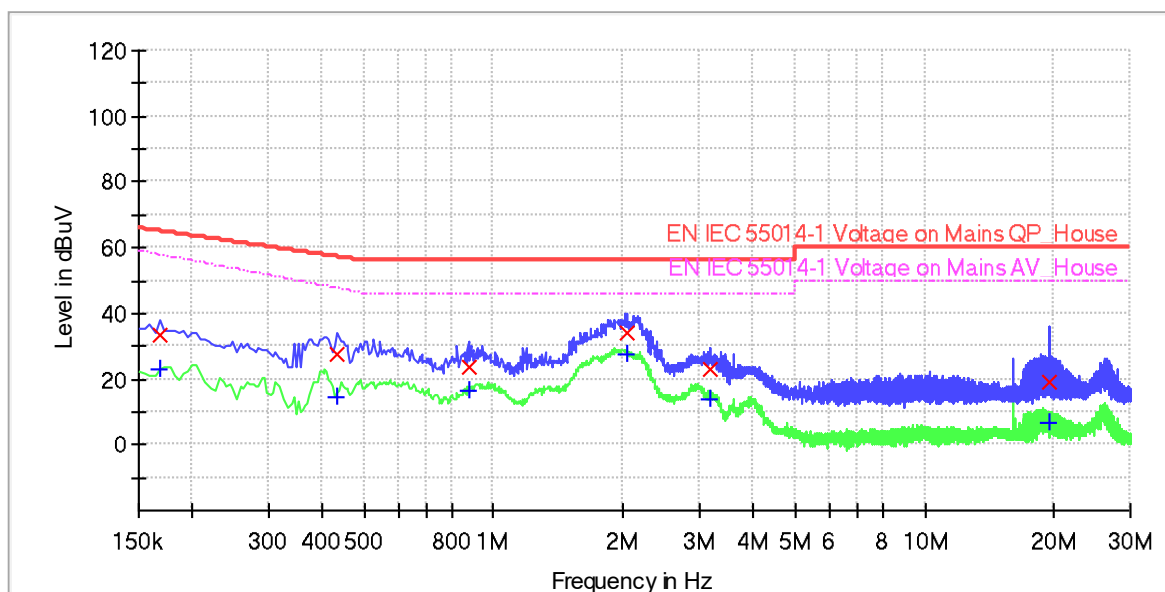
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JC26RN2/S
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, min temperature, timer on, swing, T29.5°C, H51.2%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: N
 Sample No.: NIB-987082-3

Scan Setup: ENV216_150kHz-30MHz_Preview [EMI conducted]

Hardware Setup: ENV216(Filter ON)
 Receiver: [ESR 3]
 Level Unit: dBuV

Subrange	Step Size	Detectors	IF BW	Meas. Time	Preamp
150 kHz - 30 MHz	4.5 kHz	PK+ ; AVG	9 kHz	0.01 s	0 dB



Limit and Margin

Frequency (MHz)	QuasiPeak (dBuV)	CAverage (dBuV)	Meas. Time (ms)	Bandwidth (kHz)	Line	Corr. (dB)	Margin - QPK (dB)	Limit - QPK (dBuV)	Margin - CAV (dB)	Limit - CAV (dBuV)
0.168000	33.2	22.8	1000.0	9.000	N	9.7	31.839	65.059	34.946	57.776
0.433500	27.5	14.2	1000.0	9.000	N	9.7	29.685	57.185	33.351	47.541
0.874500	23.8	16.6	1000.0	9.000	N	9.7	32.160	56.000	29.440	46.000
2.035500	34.2	27.8	1000.0	9.000	N	9.7	21.820	56.000	18.230	46.000
3.174000	22.7	14.1	1000.0	9.000	N	9.7	33.320	56.000	31.910	46.000
19.351500	19.1	7.0	1000.0	9.000	N	9.9	40.920	60.000	42.990	50.000



Test Setup

2.1.8 Test Location

This test was carried out in shielded room 120.

2.2 Discontinuous Disturbance

2.2.1 Specification Reference

EN IEC 55014-1:2021, Clause 4.4

2.2.2 Equipment Under Test

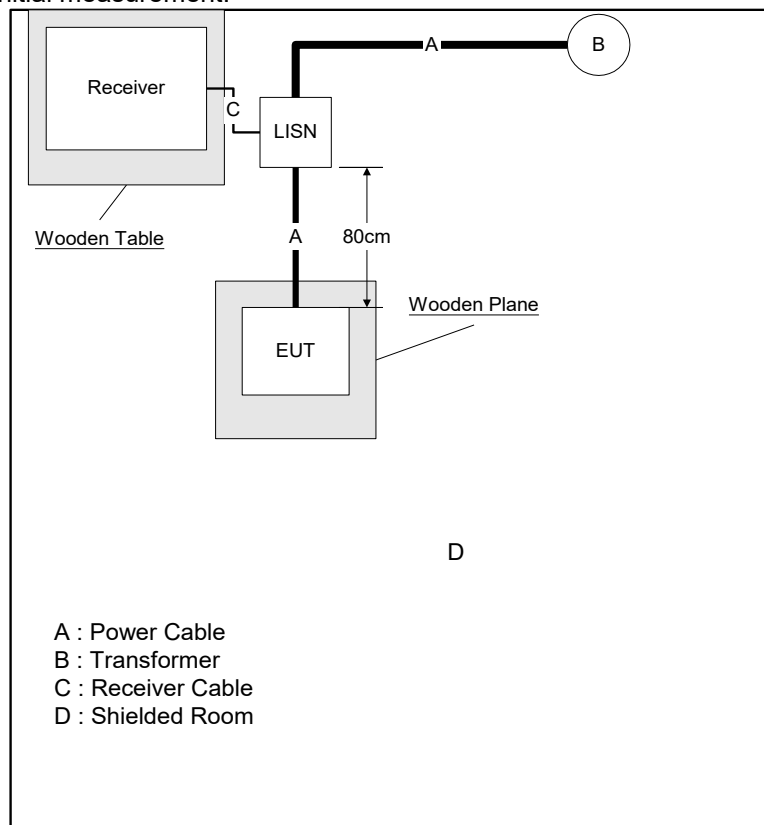
OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.2.3 Date of Test

12/18/2025, 12/25/2025, 01/27/2026

2.2.4 Test Method

All power was connected to the EUT through an Artificial Mains Network (AMN). Conducted disturbance voltage measurements on mains lines were made at the output of the AMN. An initial measurement was made in an observation period of 120 minutes, or the time to register 40 clicks. The clicks obtained was compared against the corresponding click limit L_q for the required 4 individual frequencies. The test was repeated in similar period as the initial measurement. The number of clicks during the second measurement should not exceed a quarter of that measured during the initial measurement.



2.2.5 Environmental Conditions

Ambient Temperature	10-35 °C
Relative Humidity	30-60 %
Atmospheric Pressure	1010-1040 mbar

2.2.6 Specification Limits

Frequency Range (MHz)	Click Limit Values (dB μ V)	
	N < 0.2	0.2 $\leq$ N < 30
0.15	110dB μ V	66dB μ V + 20 log (30/N)
0.5	100dB μ V	56dB μ V + 20 log (30/N)
1.4	100dB μ V	56dB μ V + 20 log (30/N)
30	104dB μ V	60dB μ V + 20 log (30/N)
Note: N refers to click rate		

2.2.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, cooling, high fan speed, min temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, cooling, high fan speed, min temperature, timer on, swing

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Discontinuous Disturbance Test

EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016PP26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, mid temperature, timer on, T29.2°C, H48.1%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Sample No.: NIB-987082-1

Test Duration:	2:00:00	LISN Phase	N	Attenuation [dB]:	25
Overload:	No				
Frequency	150 kHz	500 kHz	1.4 MHz	30 MHz	
Clicks (≤10 ms)	2	5	4	0	
Clicks (>10 ms ≤ 20 ms)	0	0	0	0	
Clicks (> 20 ms ≤ 200 ms)	0	0	1	0	
Click Rate [1/min]	0.02	0.04	0.04	0.04	
Continuous Disturbances	0	0	0	0	
L [dBuV]	66	56	56	60	
Lq [dBuV]	110	100	100	104	
Clicks > Lq	0	0	0	0	
Clicks > Lq [%]	0	0	0	0	
Max. Allowed Nr. of Clicks > Lq	-	-	-	-	
Fridge Rules	0	0	0	0	
600 ms Rule used	NO	NO	NO	NO	
Overall Correction	10	10	10	10	
Margin for PK Detector	0	0	0	0	
Result	PASSED	PASSED	PASSED	PASSED	

Discontinuous Disturbance Test

EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JA26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, mid temperature, timer on, T29.3°C, H49.2%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Sample No.: NIB-987082-2

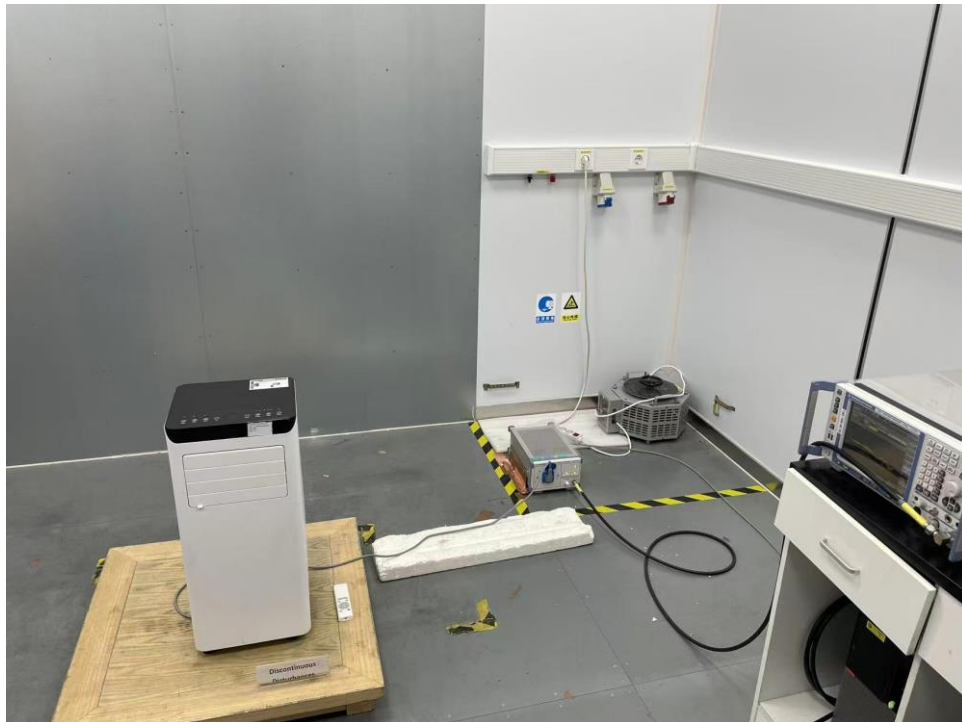
Test Duration:	2:00:00	LISN Phase:	N	Attenuation [dB]:	25
Overload:	No				
Frequency	150 kHz	500 kHz	1.4 MHz	30 MHz	
Clicks (≤10 ms)	0	1	1	0	
Clicks (>10 ms ≤ 20 ms)	0	0	0	0	
Clicks (> 20 ms ≤ 200 ms)	0	0	0	0	
Click Rate [1/min]	0.00	0.01	0.01	0.01	
Continuous Disturbances	0	0	0	0	
L [dBuV]	66	56	56	60	
Lq [dBuV]	110	100	100	104	
Clicks > Lq	0	0	0	0	
Clicks > Lq [%]	0	0	0	0	
Max. Allowed Nr. of Clicks > Lq	-	-	-	-	
Fridge Rules	0	0	0	0	
600 ms Rule used	NO	NO	NO	NO	
Overall Correction	10	10	10	10	
Margin for PK Detector	0	0	0	0	
Result	PASSED	PASSED	PASSED	PASSED	

Discontinuous Disturbance Test

EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JC26RN2/S
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, cooling, high fan speed, mid temperature, timer on, swing, T29.2°C, H50.5%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Sample No.: NIB-987082-3

Test Duration:	2:00:00	LISN Phase:	N	Attenuation [dB]:	25
Overload:	No				
Frequency	150 kHz	500 kHz	1.4 MHz	30 MHz	
Clicks (≤10 ms)	0	0	0	0	
Clicks (>10 ms ≤ 20 ms)	0	0	0	0	
Clicks (> 20 ms ≤ 200 ms)	0	0	0	0	
Click Rate [1/min]	0.00	0.00	0.00	0.00	
Continuous Disturbances	0	0	0	0	
L [dBuV]	66	56	56	60	
Lq [dBuV]	110	100	100	104	
Clicks > Lq	0	0	0	0	
Clicks > Lq [%]	0	0	0	0	
Max. Allowed Nr. of Clicks > Lq	-	-	-	-	
Fridge Rules	0	0	0	0	
600 ms Rule used	NO	NO	NO	NO	
Overall Correction	10	10	10	10	
Margin for PK Detector	0	0	0	0	
Result	PASSED	PASSED	PASSED	PASSED	



Test Setup

2.2.8 Test Location

This test was carried out in shielded room 120.

2.3 Radiated Disturbance

2.3.1 Specification Reference

EN IEC 55014-1:2021, Clause 4.3.4.5

2.3.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.3.3 Date of Test

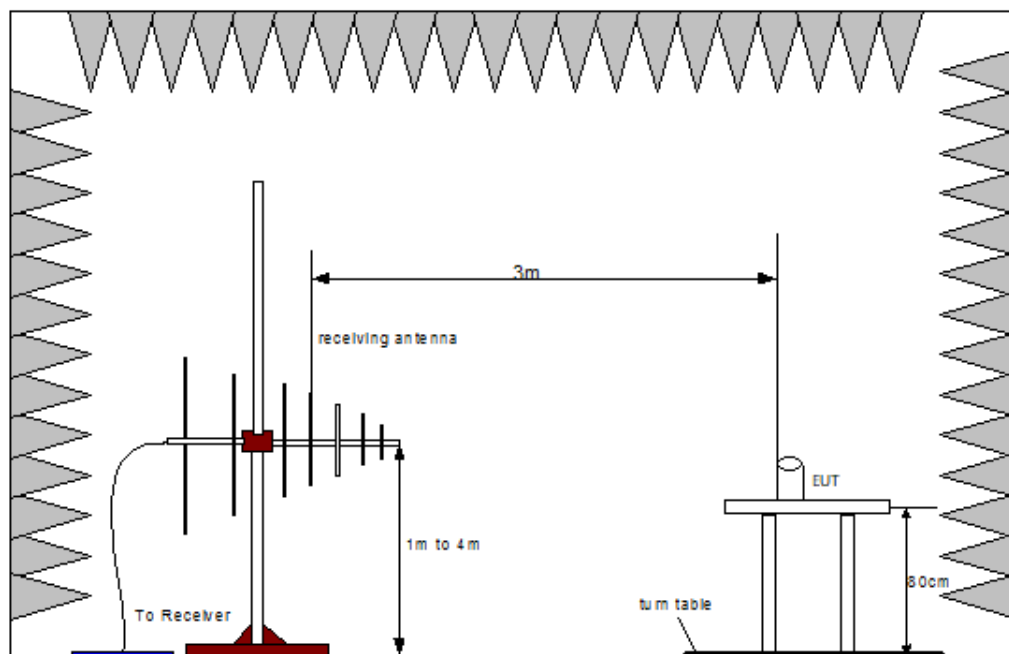
12/17/2025, 12/25/2025, 01/22/2026

2.3.4 Test Method

The EUT was set up in a semi-anechoic chamber on a remotely controlled turntable. Table top EUT was placed at at $(0,8 \pm 0,05)$ m above the reference plane of the test site selected for measurement (see Figure 16 and Figure 17 of EN IEC 55014-1: 2021).

Floor standing EUT was placed at $(0,12 \pm 0,04)$ m above the reference plane of the test site selected for measurement measurement (see Figure 19 of EN IEC 55014-1: 2021).

A prescan of the EUT emissions profile was made while varying the antenna-to-EUT azimuth and antenna-to-EUT polarization using a peak detector; measurements were taken at a 3m distance. Using the prescan list of the highest emissions detected, their bearing and associated antenna polarization, the EUT was then formally measured using a Quasi-Peak detector. The readings were maximized by adjusting the antenna height, polarization and turntable azimuth, in accordance with the specification.



2.3.5 Environmental Conditions

Ambient Temperature	10-35 °C
Relative Humidity	30-60 %
Atmospheric Pressure	1010-1040 mbar

2.3.6 Specification Limits

Radiated disturbance limits and test method for the frequency range 30MHz to 1000MHz at a measuring distance of 3 m	
Frequency range MHz	Quasi-peak limits dB(μ V/m)
30 to 230	40
230 to 1000	47

2.3.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, heating, high fan speed, max temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, heating, high fan speed, max temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Frequency Range of Test: 30 MHz to 1 GHz

30-1000MHz Radiated Emission

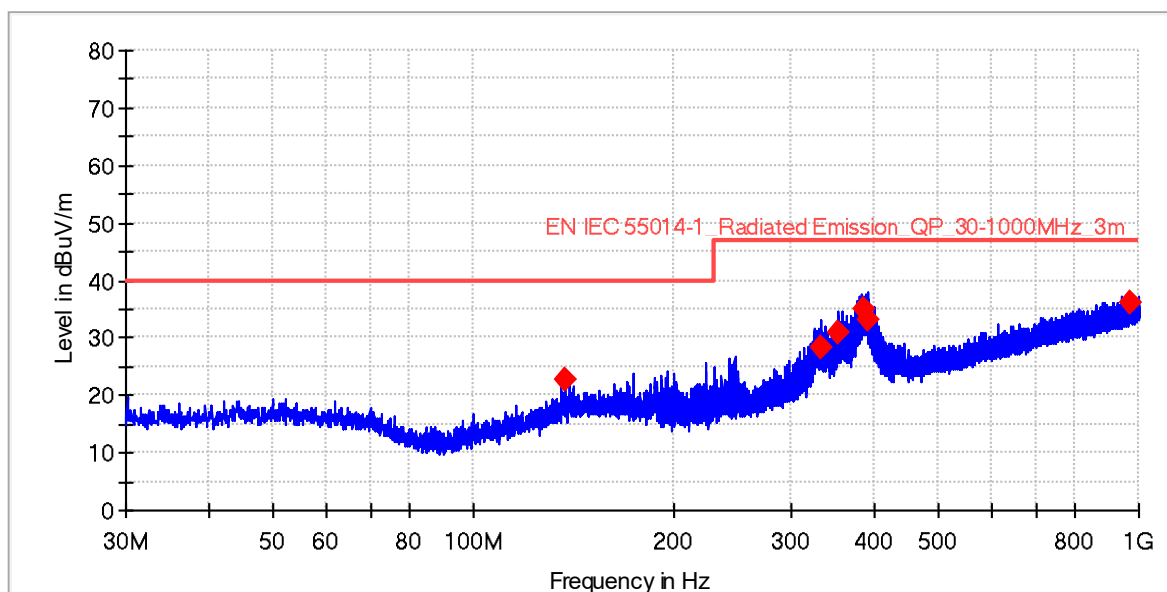
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016PP26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on,
 T17.3°C, H50.2%, P102.0kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Horizontal
 Sample No: NIB-987082-1

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
137.473020	22.62	40.00	17.38	1000.0	120.000	200.0	H	103.0	21.2
332.558621	28.18	47.00	18.82	1000.0	120.000	100.0	H	179.0	24.4
354.121768	30.80	47.00	16.20	1000.0	120.000	100.0	H	200.0	24.7
384.460269	34.80	47.00	12.20	1000.0	120.000	100.0	H	148.0	25.8
390.921531	33.02	47.00	13.98	1000.0	120.000	100.0	H	141.0	26.0
968.551584	36.24	47.00	10.76	1000.0	120.000	100.0	H	298.0	37.2

30-1000MHz Radiated Emission

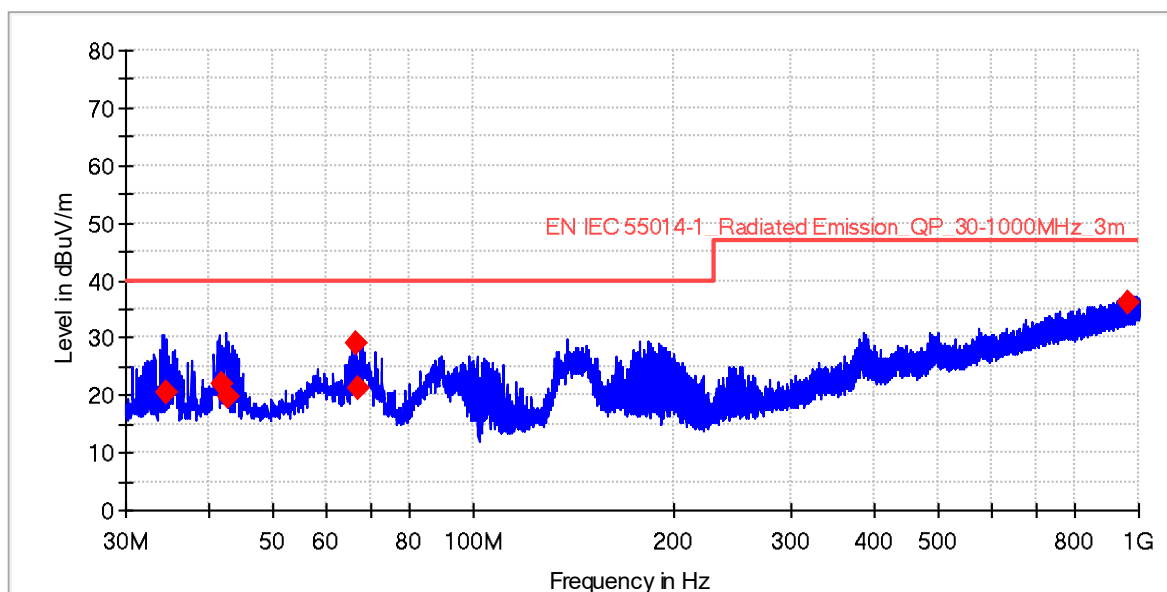
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016PP26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on,
 T17.3°C, H50.2%, P102.0kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Vertical
 Sample No: NIB-987082-1

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
34.523789	20.47	40.00	19.53	1000.0	120.000	100.0	V	155.0	19.9
41.955386	21.87	40.00	18.13	1000.0	120.000	100.0	V	134.0	20.8
42.673881	19.63	40.00	20.37	1000.0	120.000	200.0	V	358.0	20.8
66.435718	28.86	40.00	11.14	1000.0	120.000	100.0	V	155.0	20.1
67.274365	21.28	40.00	18.72	1000.0	120.000	100.0	V	120.0	20.0
961.376000	36.21	47.00	10.79	1000.0	120.000	100.0	V	274.0	37.2

30-1000MHz Radiated Emission

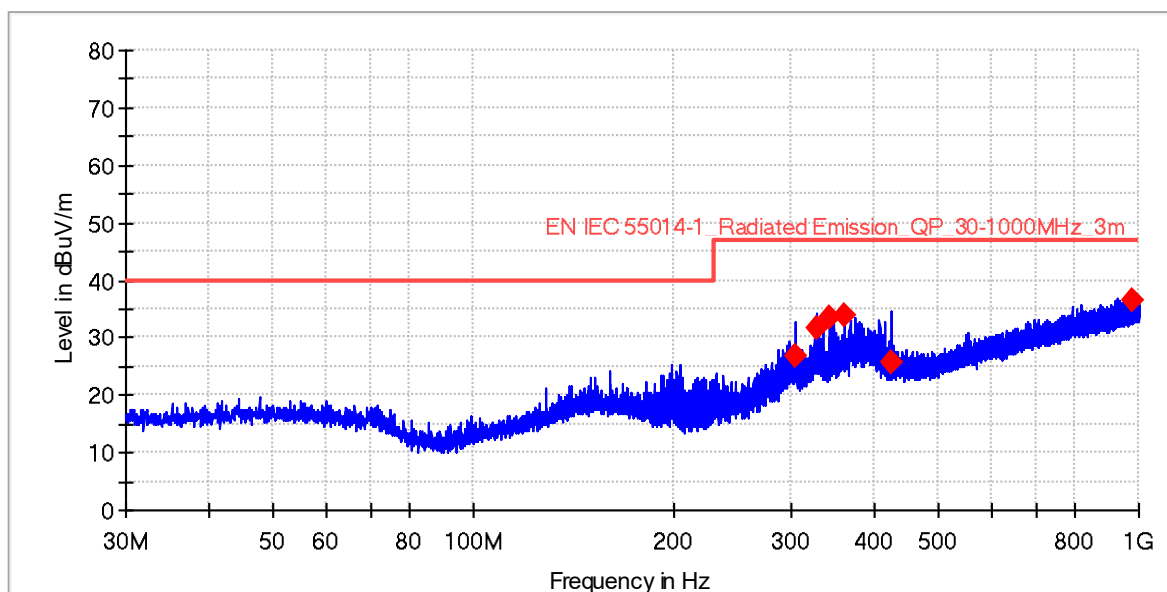
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JA26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on,
 T17.5°C, H50.1%, P102.0kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Horizontal
 Sample No: NIB-987082-2

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
303.934959	26.76	47.00	20.24	1000.0	120.000	100.0	H	186.0	23.4
327.844707	31.77	47.00	15.23	1000.0	120.000	100.0	H	152.0	24.3
343.881733	33.39	47.00	13.61	1000.0	120.000	100.0	H	165.0	24.5
359.918759	33.95	47.00	13.05	1000.0	120.000	100.0	H	124.0	24.8
423.691769	25.72	47.00	21.28	1000.0	120.000	200.0	H	190.0	27.0
980.348870	36.43	47.00	10.57	1000.0	120.000	100.0	H	255.0	37.3

30-1000MHz Radiated Emission

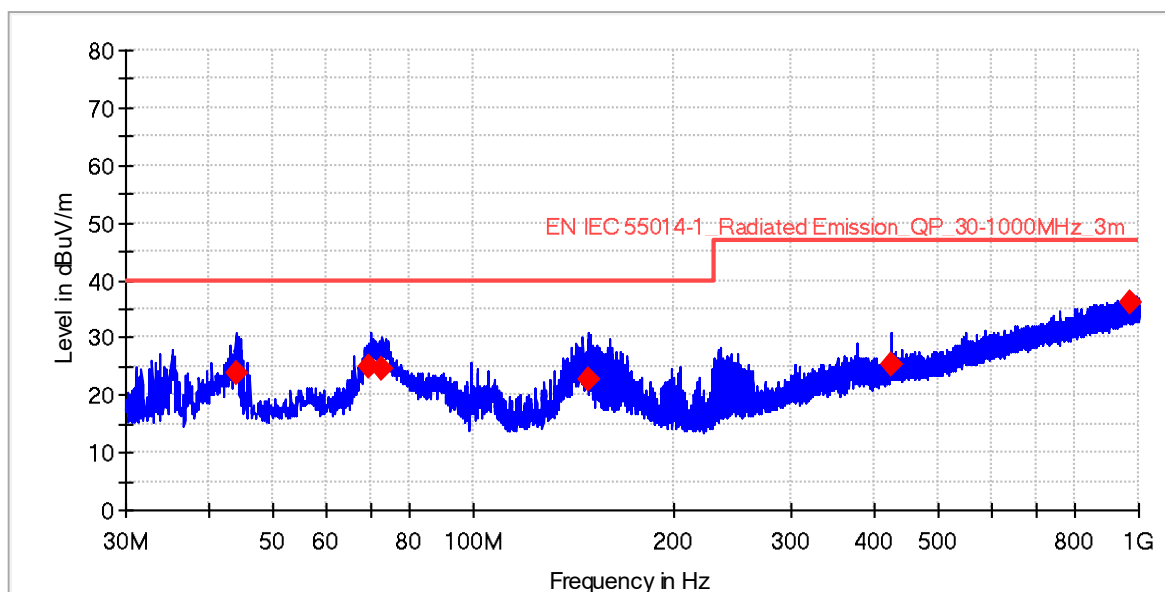
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JA26RN2
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on, T17.5°C, H50.1%, P102.0kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Vertical
 Sample No: NIB-987082-2

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
44.194170	23.96	40.00	16.04	1000.0	120.000	100.0	V	179.0	21.0
69.558000	24.79	40.00	15.21	1000.0	120.000	100.0	V	179.0	19.6
72.782322	24.42	40.00	15.58	1000.0	120.000	100.0	V	214.0	18.8
148.972650	22.65	40.00	17.35	1000.0	120.000	100.0	V	158.0	22.0
423.317000	25.14	47.00	21.86	1000.0	120.000	100.0	V	74.0	27.0
967.765500	36.17	47.00	10.83	1000.0	120.000	100.0	V	342.0	37.2

30-1000MHz Radiated Emission

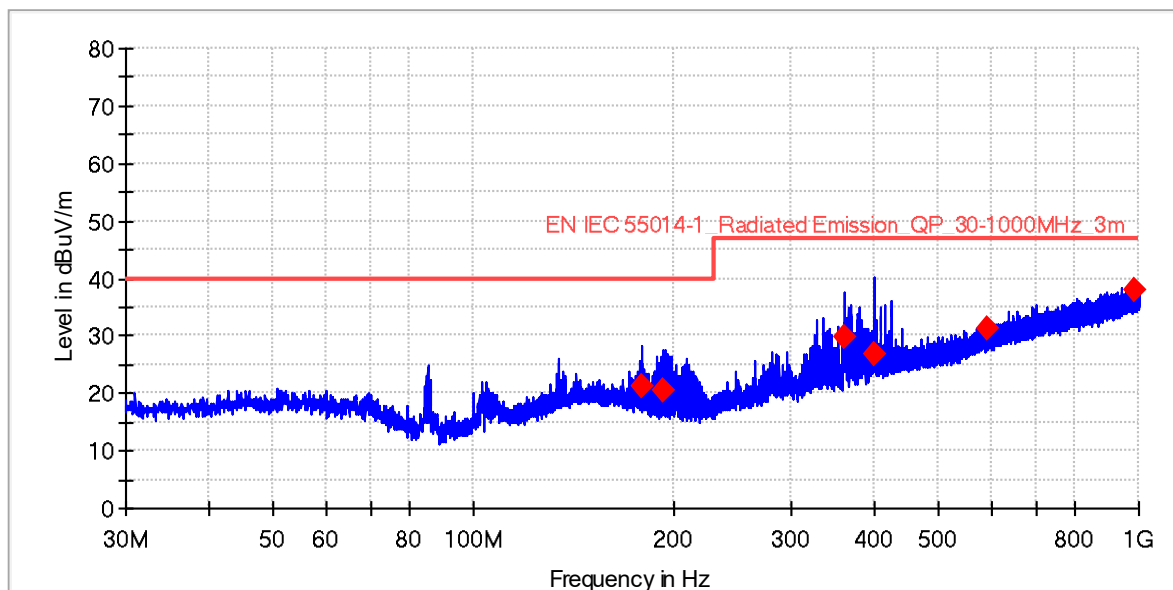
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JC26RN2/S
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on, swing, T16.9°C, H49.3%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Horizontal
 Sample No: NIB-987082-3

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
179.055877	21.21	40.00	18.79	1000.0	120.000	200.0	H	223.0	20.8
193.426233	20.57	40.00	19.43	1000.0	120.000	200.0	H	216.0	19.4
360.356133	29.67	47.00	17.33	1000.0	120.000	100.0	H	175.0	24.9
399.552000	26.76	47.00	20.24	1000.0	120.000	100.0	H	224.0	26.2
593.685000	31.42	47.00	15.58	1000.0	120.000	100.0	H	259.0	31.3
983.686000	38.11	47.00	8.89	1000.0	120.000	100.0	H	63.0	37.2

30-1000MHz Radiated Emission

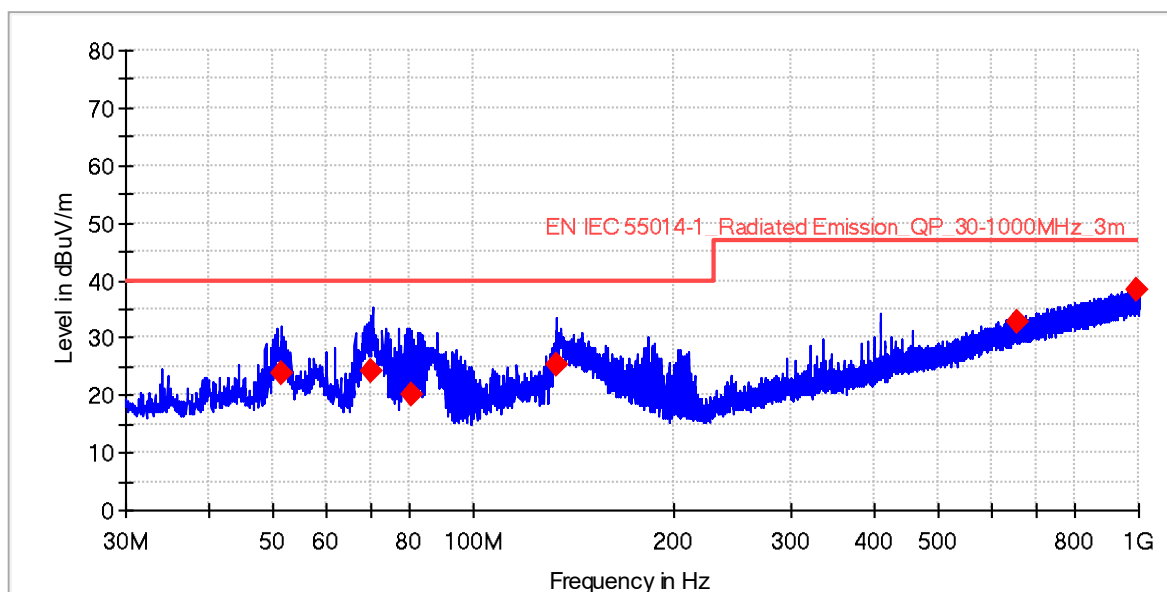
EUT Information

EUT Name: Portable air conditioner
 Model: OL-A016JC26RN2/S
 Client: Zhejiang Oulun Electric Co., Ltd.
 Op Cond: Power on, heating, high fan speed, max temperature, timer on, swing, T16.9°C, H49.3%, P102.1kPa
 Operator: Peiqiang WANG
 Standard: EN IEC 55014-1
 Comment: Vertical
 Sample No: NIB-987082-3

Sweep Setup: RE_30MHz-1GHz_Pre [EMI radiated]

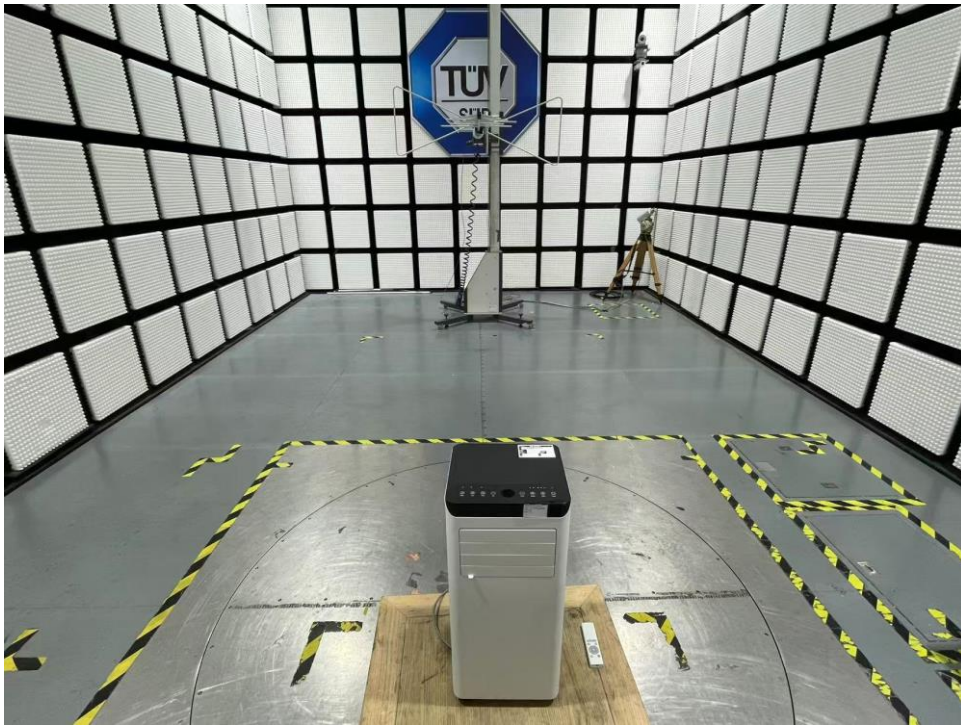
Hardware Setup: RE_30MHz-1GHz
 Receiver: [ESR 7]
 Level Unit: dBuV/m

Subrange	Step Size	Detectors	Bandwidth	Sweep Time	Preamp
30 MHz - 1 GHz	48.5 kHz	PK+	120 kHz	0.2 s	20 dB



Final Result

Frequency (MHz)	QuasiPeak (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB/m)
51.249689	24.00	40.00	16.00	1000.0	120.000	100.0	V	283.0	21.4
69.870301	24.30	40.00	15.70	1000.0	120.000	100.0	V	30.0	19.6
80.491096	19.92	40.00	20.08	1000.0	120.000	100.0	V	184.0	16.8
133.369853	25.43	40.00	14.57	1000.0	120.000	100.0	V	64.0	20.9
656.190535	32.68	47.00	14.32	1000.0	120.000	100.0	V	276.0	32.4
994.136644	38.46	47.00	8.54	1000.0	120.000	100.0	V	212.0	37.5



Test Setup

2.3.8 Test Location

This test was carried out in 3-meter semi-anechoic chamber.

2.4 Harmonic current emission

2.4.1 Specification Reference

EN IEC 61000-3-2:2019/A2:2024, Clause 7

2.4.2 Equipment Under Test

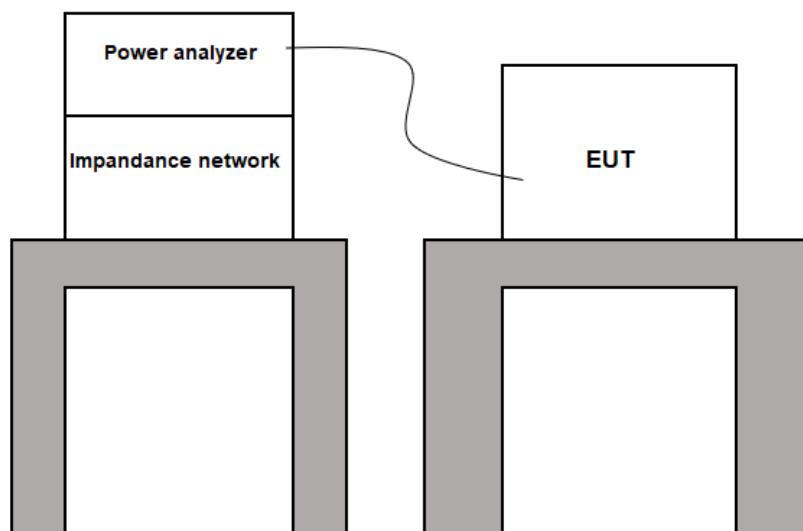
OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.4.3 Date of Test

12/19/2025, 12/23/2025, 01/29/2026

2.4.4 Test Method

Harmonic components shall be measured in accordance with the requirements given in Annex A of EN IEC 61000-3-2 for the test circuit and the supply source.



2.4.5 Environmental Conditions

Ambient Temperature	10-35 °C
Relative Humidity	30-60 %
Atmospheric Pressure	1010-1040 mbar

2.4.6 Specification Limits

Limits for class A Equipment	
Harmonic order h	Maximum permissible harmonic current A
Odd harmonics	
3	2.30
5	1.14
7	0.77
9	0.40
11	0.33
13	0.21
$15 \leq h \leq 39$	$0.15(15/h)$
Even harmonics	
2	1.08
4	0.43
6	0.30
$8 \leq h \leq 40$	$0.23(8/h)$

2.4.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2: AC Powered/Power on, cooling, high fan speed, min temperature, timer on;

OL-A016JA26RN2: AC Powered/Power on, dehumidifier, high fan speed, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, heating, high fan speed, max temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016PP26RN2 Power on, cooling, high fan speed, min temperature, timer on

Test Standard : IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Tester : Peiqiang WANG

Test Date : 12/19/2025 3:14:30 PM

Climatic Conditions

Temperature : 27 °C Pressure : 103 kPa Humidity : 58 %

Standard Group: Industry

Standard Name: IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Device Under Test: **PASS**

Power Source: **PASS**

Connection Type: L - N

Classification: Class A

Appli. of Limits: less than or equal to 150 % (Without POHC Enhancement)

Check Harmonics 2..40

First detected harmonic order > 150 %

Line 1: **None**

Harmonics orders > 150 %

Line 1: **None**

Harmonics orders with average > 100 %

Line 1: **None**

Measured values

Fundamental Current

Line 1: 3.552 A

Active input Power

Line 1: 808.721 W *

Circuit power factor

Line 1: 0.968 *

* Absolute value.

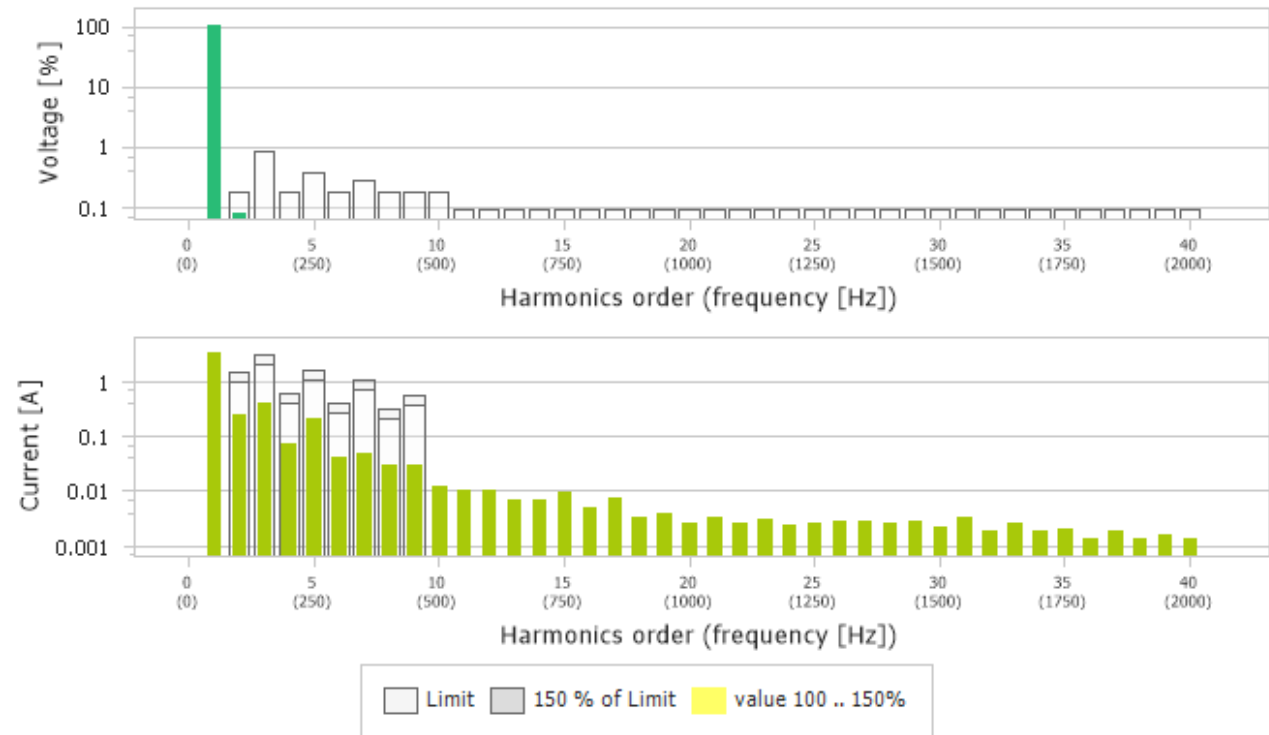
Current Test Result

Average and Maximum harmonic current results									
Hn	Average				Maximum				Harmonic Result
	I _{eff} [A]	of Limit [%]	Limit [A]	Result	I _{eff} [A]	of Limit [%]	Limit [A]	Result	
1	3.535				3.552				
2	0.275	25.452	1.080	PASS	0.277	17.079	1.620	PASS	PASS
3	0.452	19.631	2.300	PASS	0.454	13.165	3.450	PASS	PASS
4	0.075	17.426	0.430	PASS	0.077	11.891	0.645	PASS	PASS
5	0.225	19.729	1.140	PASS	0.228	13.357	1.710	PASS	PASS
6	0.046	15.182	0.300	PASS	0.046	10.320	0.450	PASS	PASS
7	0.050	6.514	0.770	PASS	0.052	4.493	1.155	PASS	PASS
8	0.031	13.402	0.230	PASS	0.031	9.063	0.345	PASS	PASS
9	0.031	7.704	0.400	PASS	0.032	5.288	0.600	PASS	PASS
10	0.013	6.955	0.184	n/a	0.013	4.802	0.276	n/a	PASS
11	0.011	3.339	0.330	n/a	0.012	2.358	0.495	n/a	PASS
12	0.011	7.292	0.153	n/a	0.012	5.026	0.230	n/a	PASS
13	0.007	3.322	0.210	n/a	0.008	2.393	0.315	n/a	PASS
14	0.007	5.341	0.131	n/a	0.007	3.695	0.197	n/a	PASS
15	0.010	6.556	0.150	n/a	0.010	4.540	0.225	n/a	PASS
16	0.005	4.641	0.115	n/a	0.006	3.228	0.173	n/a	PASS
17	0.008	6.044	0.132	n/a	0.008	4.184	0.199	n/a	PASS
18	0.003	3.203	0.102	n/a	0.004	2.364	0.153	n/a	PASS
19	0.004	3.305	0.118	n/a	0.004	2.379	0.178	n/a	PASS
20	0.003	2.761	0.092	n/a	0.003	2.060	0.138	n/a	PASS
21	0.003	3.122	0.107	n/a	0.004	2.281	0.161	n/a	PASS
22	0.002	2.852	0.084	n/a	0.003	2.184	0.125	n/a	PASS
23	0.003	3.197	0.098	n/a	0.003	2.342	0.147	n/a	PASS
24	0.002	2.950	0.077	n/a	0.003	2.207	0.115	n/a	PASS
25	0.003	2.980	0.090	n/a	0.003	2.144	0.135	n/a	PASS
26	0.003	3.933	0.071	n/a	0.003	2.832	0.106	n/a	PASS
27	0.003	3.255	0.083	n/a	0.003	2.399	0.125	n/a	PASS
28	0.003	3.844	0.066	n/a	0.003	2.748	0.099	n/a	PASS
29	0.003	3.372	0.078	n/a	0.003	2.534	0.116	n/a	PASS
30	0.002	3.369	0.061	n/a	0.002	2.570	0.092	n/a	PASS
31	0.003	4.098	0.073	n/a	0.004	3.300	0.109	n/a	PASS
32	0.002	3.186	0.058	n/a	0.002	2.336	0.086	n/a	PASS
33	0.002	3.433	0.068	n/a	0.003	2.725	0.102	n/a	PASS
34	0.002	3.333	0.054	n/a	0.002	2.537	0.081	n/a	PASS
35	0.002	2.941	0.064	n/a	0.002	2.219	0.096	n/a	PASS
36	0.001	2.546	0.051	n/a	0.001	1.930	0.077	n/a	PASS
37	0.002	2.899	0.061	n/a	0.002	2.223	0.091	n/a	PASS
38	0.001	2.564	0.048	n/a	0.001	1.948	0.073	n/a	PASS
39	0.002	2.616	0.058	n/a	0.002	1.917	0.087	n/a	PASS
40	0.001	2.751	0.046	n/a	0.001	2.064	0.069	n/a	PASS

Note: Harmonic currents less than 0.6 % of the input current measured under the test conditions, or less than 5 mA, whichever is greater, are disregarded.

Voltage Source Verification

Harmonic voltage results				
Hn	Ueff [V]	Ueff [%]	Limit [%]	Result
1	232.250	100.978		
2	0.206	0.089	0.200	PASS
3	0.038	0.017	0.900	PASS
4	0.077	0.034	0.200	PASS
5	0.048	0.021	0.400	PASS
6	0.049	0.021	0.200	PASS
7	0.049	0.021	0.300	PASS
8	0.038	0.016	0.200	PASS
9	0.043	0.019	0.200	PASS
10	0.031	0.014	0.200	PASS
11	0.028	0.012	0.100	PASS
12	0.029	0.013	0.100	PASS
13	0.030	0.013	0.100	PASS
14	0.023	0.010	0.100	PASS
15	0.027	0.012	0.100	PASS
16	0.020	0.009	0.100	PASS
17	0.019	0.008	0.100	PASS
18	0.016	0.007	0.100	PASS
19	0.017	0.007	0.100	PASS
20	0.017	0.007	0.100	PASS
21	0.013	0.006	0.100	PASS
22	0.018	0.008	0.100	PASS
23	0.017	0.007	0.100	PASS
24	0.014	0.006	0.100	PASS
25	0.017	0.007	0.100	PASS
26	0.012	0.005	0.100	PASS
27	0.023	0.010	0.100	PASS
28	0.017	0.007	0.100	PASS
29	0.019	0.008	0.100	PASS
30	0.020	0.009	0.100	PASS
31	0.013	0.006	0.100	PASS
32	0.014	0.006	0.100	PASS
33	0.018	0.008	0.100	PASS
34	0.009	0.004	0.100	PASS
35	0.014	0.006	0.100	PASS
36	0.009	0.004	0.100	PASS
37	0.012	0.005	0.100	PASS
38	0.012	0.005	0.100	PASS
39	0.016	0.007	0.100	PASS
40	0.013	0.006	0.100	PASS



Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016JA26RN2 Power on, dehumidifier, high fan speed, timer on

Test Standard : IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Tester : Peiqiang WANG

Test Date : 12/23/2025 9:39:48 AM

Climatic Conditions

Temperature : 24 °C Pressure : 103 kPa Humidity : 58 %

Standard Group: Industry

Standard Name: IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Device Under Test: **PASS**

Power Source: **PASS**

Connection Type: L - N

Classification: Class A

Appli. of Limits: less than or equal to 150 % (Without POHC Enhancement)

Check Harmonics 2..40

First detected harmonic order > 150 %

Line 1: **None**

Harmonics orders > 150 %

Line 1: **None**

Harmonics orders with average > 100 %

Line 1: **None**

Measured values

Fundamental Current

Line 1: 3.805 A

Active input Power

Line 1: 870.829 W *

Circuit power factor

Line 1: 0.975 *

* Absolute value.

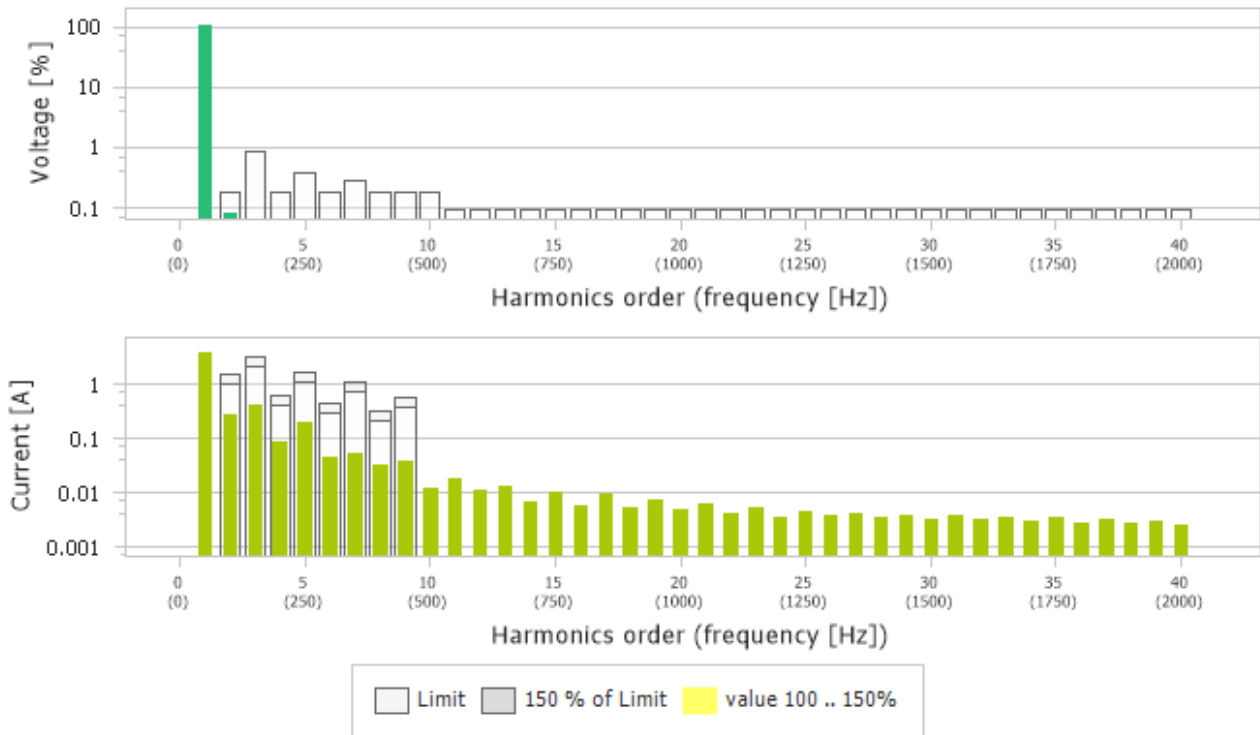
Current Test Result

Average and Maximum harmonic current results									
Hn	Average				Maximum				Harmonic Result
	I _{eff} [A]	of Limit [%]	Limit [A]	Result	I _{eff} [A]	of Limit [%]	Limit [A]	Result	
1	2.648				3.805				
2	0.186	17.195	1.080	PASS	0.291	17.953	1.620	PASS	PASS
3	0.292	12.691	2.300	PASS	0.444	12.864	3.450	PASS	PASS
4	0.056	13.019	0.430	PASS	0.090	13.984	0.645	PASS	PASS
5	0.133	11.627	1.140	PASS	0.206	12.024	1.710	PASS	PASS
6	0.030	9.972	0.300	PASS	0.046	10.312	0.450	PASS	PASS
7	0.037	4.848	0.770	PASS	0.055	4.759	1.155	PASS	PASS
8	0.022	9.665	0.230	n/a	0.034	9.986	0.345	PASS	PASS
9	0.024	6.091	0.400	PASS	0.039	6.431	0.600	PASS	PASS
10	0.008	4.478	0.184	n/a	0.012	4.456	0.276	n/a	PASS
11	0.010	2.911	0.330	n/a	0.019	3.882	0.495	n/a	PASS
12	0.008	5.057	0.153	n/a	0.011	5.000	0.230	n/a	PASS
13	0.009	4.086	0.210	n/a	0.014	4.316	0.315	n/a	PASS
14	0.005	3.615	0.131	n/a	0.007	3.558	0.197	n/a	PASS
15	0.009	6.014	0.150	n/a	0.011	4.836	0.225	n/a	PASS
16	0.004	3.354	0.115	n/a	0.006	3.344	0.173	n/a	PASS
17	0.007	5.014	0.132	n/a	0.010	4.877	0.199	n/a	PASS
18	0.003	2.544	0.102	n/a	0.005	3.466	0.153	n/a	PASS
19	0.004	3.248	0.118	n/a	0.008	4.424	0.178	n/a	PASS
20	0.002	2.372	0.092	n/a	0.005	3.560	0.138	n/a	PASS
21	0.003	3.173	0.107	n/a	0.006	3.905	0.161	n/a	PASS
22	0.002	2.351	0.084	n/a	0.004	3.543	0.125	n/a	PASS
23	0.003	3.068	0.098	n/a	0.005	3.596	0.147	n/a	PASS
24	0.002	2.374	0.077	n/a	0.004	3.284	0.115	n/a	PASS
25	0.002	2.777	0.090	n/a	0.005	3.580	0.135	n/a	PASS
26	0.002	2.956	0.071	n/a	0.004	3.607	0.106	n/a	PASS
27	0.003	3.197	0.083	n/a	0.004	3.358	0.125	n/a	PASS
28	0.002	2.912	0.066	n/a	0.003	3.531	0.099	n/a	PASS
29	0.003	3.271	0.078	n/a	0.004	3.433	0.116	n/a	PASS
30	0.002	2.817	0.061	n/a	0.003	3.486	0.092	n/a	PASS
31	0.003	3.602	0.073	n/a	0.004	3.577	0.109	n/a	PASS
32	0.002	2.684	0.058	n/a	0.003	3.816	0.086	n/a	PASS
33	0.002	3.400	0.068	n/a	0.004	3.534	0.102	n/a	PASS
34	0.001	2.758	0.054	n/a	0.003	3.719	0.081	n/a	PASS
35	0.002	2.875	0.064	n/a	0.004	3.634	0.096	n/a	PASS
36	0.001	2.373	0.051	n/a	0.003	3.800	0.077	n/a	PASS
37	0.002	2.643	0.061	n/a	0.003	3.524	0.091	n/a	PASS
38	0.001	2.400	0.048	n/a	0.003	3.785	0.073	n/a	PASS
39	0.001	2.442	0.058	n/a	0.003	3.495	0.087	n/a	PASS
40	0.001	2.516	0.046	n/a	0.003	3.781	0.069	n/a	PASS

Note: Harmonic currents less than 0.6 % of the input current measured under the test conditions, or less than 5 mA, whichever is greater, are disregarded.

Voltage Source Verification

Harmonic voltage results				
Hn	Ueff [V]	Ueff [%]	Limit [%]	Result
1	232.497	101.086		
2	0.206	0.090	0.200	PASS
3	0.051	0.022	0.900	PASS
4	0.076	0.033	0.200	PASS
5	0.054	0.023	0.400	PASS
6	0.062	0.027	0.200	PASS
7	0.047	0.020	0.300	PASS
8	0.040	0.018	0.200	PASS
9	0.043	0.019	0.200	PASS
10	0.030	0.013	0.200	PASS
11	0.031	0.013	0.100	PASS
12	0.029	0.013	0.100	PASS
13	0.030	0.013	0.100	PASS
14	0.025	0.011	0.100	PASS
15	0.030	0.013	0.100	PASS
16	0.022	0.009	0.100	PASS
17	0.018	0.008	0.100	PASS
18	0.017	0.007	0.100	PASS
19	0.018	0.008	0.100	PASS
20	0.017	0.007	0.100	PASS
21	0.015	0.007	0.100	PASS
22	0.018	0.008	0.100	PASS
23	0.016	0.007	0.100	PASS
24	0.013	0.006	0.100	PASS
25	0.016	0.007	0.100	PASS
26	0.013	0.005	0.100	PASS
27	0.024	0.011	0.100	PASS
28	0.014	0.006	0.100	PASS
29	0.017	0.007	0.100	PASS
30	0.018	0.008	0.100	PASS
31	0.014	0.006	0.100	PASS
32	0.014	0.006	0.100	PASS
33	0.017	0.008	0.100	PASS
34	0.009	0.004	0.100	PASS
35	0.012	0.005	0.100	PASS
36	0.009	0.004	0.100	PASS
37	0.011	0.005	0.100	PASS
38	0.014	0.006	0.100	PASS
39	0.018	0.008	0.100	PASS
40	0.014	0.006	0.100	PASS



Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016JC26RN2/S Power on, heating, high fan speed, max temperature, timer on, swing

Test Standard : IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Tester : Peiqiang WANG

Test Date : 1/29/2026 9:48:36 AM

Climatic Conditions

Temperature : 18 °C Pressure : 103 kPa Humidity : 58 %

Standard Group: Industry

Standard Name: IEC 61000-3-2 (Edition 5.1)
Limits for harmonic current emissions (equipment input current < 16 A per phase)

Device Under Test: **PASS**

Power Source: **PASS**

Connection Type: L - N

Classification: Class A

Appli. of Limits: less than or equal to 150 % (Without POHC Enhancement)

Check Harmonics 2..40

First detected harmonic order > 150 %

Line 1: **None**

Harmonics orders > 150 %

Line 1: **None**

Harmonics orders with average > 100 %

Line 1: **None**

Measured values

Fundamental Current

Line 1: 3.642 A

Active input Power

Line 1: 840.859 W *

Circuit power factor

Line 1: 0.991 *

* Absolute value.

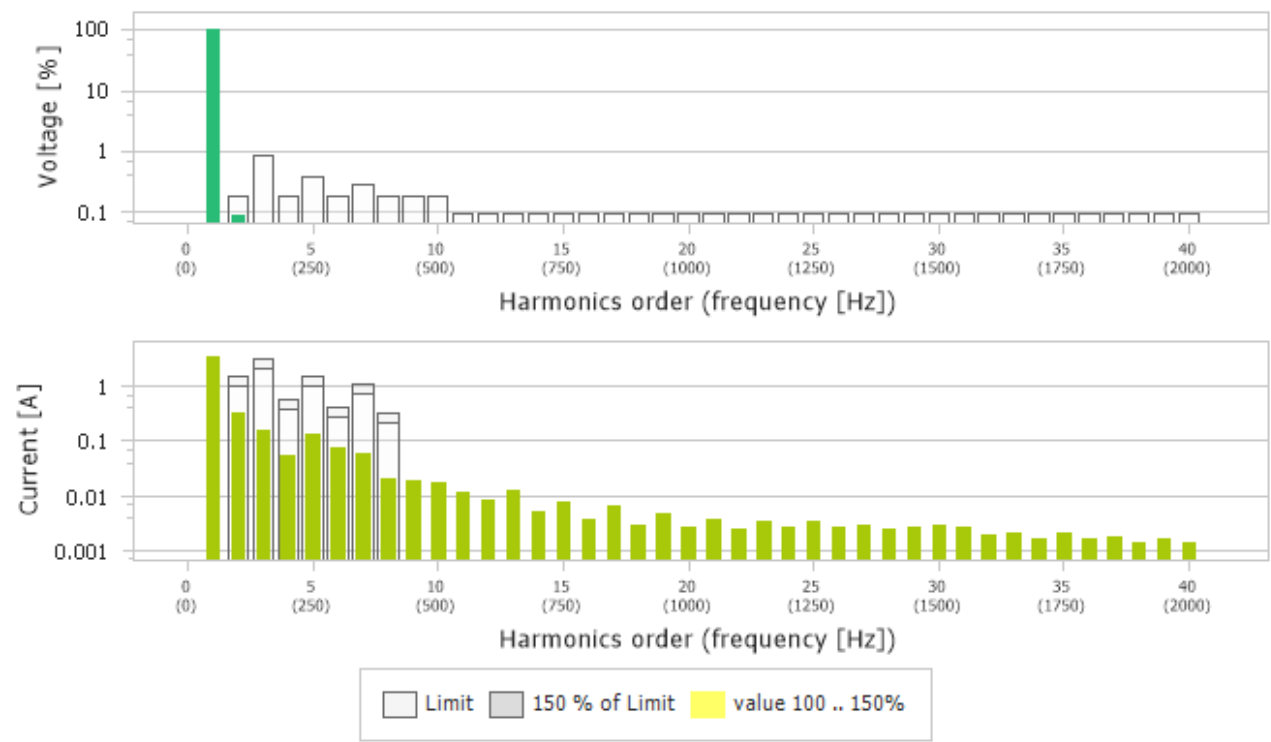
Current Test Result

Average and Maximum harmonic current results									
Hn	Average				Maximum				Harmonic Result
	I _{eff} [A]	of Limit [%]	Limit [A]	Result	I _{eff} [A]	of Limit [%]	Limit [A]	Result	
1	3.548				3.642				
2	0.358	33.131	1.080	PASS	0.365	22.544	1.620	PASS	PASS
3	0.167	7.258	2.300	PASS	0.173	5.002	3.450	PASS	PASS
4	0.056	13.001	0.430	PASS	0.058	9.061	0.645	PASS	PASS
5	0.128	11.209	1.140	PASS	0.140	8.189	1.710	PASS	PASS
6	0.077	25.600	0.300	PASS	0.079	17.570	0.450	PASS	PASS
7	0.056	7.314	0.770	PASS	0.062	5.379	1.155	PASS	PASS
8	0.019	8.169	0.230	n/a	0.022	6.440	0.345	PASS	PASS
9	0.019	4.701	0.400	n/a	0.021	3.471	0.600	n/a	PASS
10	0.019	10.181	0.184	n/a	0.019	6.912	0.276	n/a	PASS
11	0.012	3.674	0.330	n/a	0.013	2.563	0.495	n/a	PASS
12	0.008	5.048	0.153	n/a	0.009	3.815	0.230	n/a	PASS
13	0.012	5.763	0.210	n/a	0.013	4.175	0.315	n/a	PASS
14	0.005	3.895	0.131	n/a	0.005	2.757	0.197	n/a	PASS
15	0.008	5.216	0.150	n/a	0.008	3.714	0.225	n/a	PASS
16	0.004	3.299	0.115	n/a	0.004	2.347	0.173	n/a	PASS
17	0.007	5.140	0.132	n/a	0.007	3.531	0.199	n/a	PASS
18	0.003	2.834	0.102	n/a	0.003	2.040	0.153	n/a	PASS
19	0.005	3.911	0.118	n/a	0.005	2.730	0.178	n/a	PASS
20	0.003	2.841	0.092	n/a	0.003	2.108	0.138	n/a	PASS
21	0.004	3.573	0.107	n/a	0.004	2.528	0.161	n/a	PASS
22	0.002	2.929	0.084	n/a	0.003	2.102	0.125	n/a	PASS
23	0.003	3.357	0.098	n/a	0.003	2.357	0.147	n/a	PASS
24	0.003	3.331	0.077	n/a	0.003	2.477	0.115	n/a	PASS
25	0.004	3.935	0.090	n/a	0.004	2.753	0.135	n/a	PASS
26	0.003	3.603	0.071	n/a	0.003	2.639	0.106	n/a	PASS
27	0.003	3.533	0.083	n/a	0.003	2.499	0.125	n/a	PASS
28	0.003	3.813	0.066	n/a	0.003	2.730	0.099	n/a	PASS
29	0.003	3.378	0.078	n/a	0.003	2.479	0.116	n/a	PASS
30	0.003	4.459	0.061	n/a	0.003	3.211	0.092	n/a	PASS
31	0.003	3.586	0.073	n/a	0.003	2.574	0.109	n/a	PASS
32	0.002	3.277	0.058	n/a	0.002	2.395	0.086	n/a	PASS
33	0.002	2.998	0.068	n/a	0.002	2.119	0.102	n/a	PASS
34	0.002	3.021	0.054	n/a	0.002	2.191	0.081	n/a	PASS
35	0.002	3.045	0.064	n/a	0.002	2.208	0.096	n/a	PASS
36	0.002	2.937	0.051	n/a	0.002	2.160	0.077	n/a	PASS
37	0.002	2.914	0.061	n/a	0.002	2.086	0.091	n/a	PASS
38	0.001	2.854	0.048	n/a	0.002	2.068	0.073	n/a	PASS
39	0.002	2.775	0.058	n/a	0.002	1.994	0.087	n/a	PASS
40	0.001	2.966	0.046	n/a	0.002	2.200	0.069	n/a	PASS

Note: Harmonic currents less than 0.6 % of the input current measured under the test conditions, or less than 5 mA, whichever is greater, are disregarded.

Voltage Source Verification

Harmonic voltage results				
Hn	Ueff [V]	Ueff [%]	Limit [%]	Result
1	231.443	100.627		
2	0.211	0.092	0.200	PASS
3	0.039	0.017	0.900	PASS
4	0.074	0.032	0.200	PASS
5	0.057	0.025	0.400	PASS
6	0.055	0.024	0.200	PASS
7	0.043	0.019	0.300	PASS
8	0.047	0.020	0.200	PASS
9	0.048	0.021	0.200	PASS
10	0.035	0.015	0.200	PASS
11	0.034	0.015	0.100	PASS
12	0.026	0.011	0.100	PASS
13	0.029	0.012	0.100	PASS
14	0.023	0.010	0.100	PASS
15	0.033	0.014	0.100	PASS
16	0.023	0.010	0.100	PASS
17	0.021	0.009	0.100	PASS
18	0.020	0.009	0.100	PASS
19	0.018	0.008	0.100	PASS
20	0.017	0.008	0.100	PASS
21	0.018	0.008	0.100	PASS
22	0.017	0.007	0.100	PASS
23	0.016	0.007	0.100	PASS
24	0.013	0.006	0.100	PASS
25	0.016	0.007	0.100	PASS
26	0.014	0.006	0.100	PASS
27	0.027	0.012	0.100	PASS
28	0.015	0.007	0.100	PASS
29	0.018	0.008	0.100	PASS
30	0.018	0.008	0.100	PASS
31	0.012	0.005	0.100	PASS
32	0.014	0.006	0.100	PASS
33	0.022	0.010	0.100	PASS
34	0.009	0.004	0.100	PASS
35	0.014	0.006	0.100	PASS
36	0.010	0.004	0.100	PASS
37	0.011	0.005	0.100	PASS
38	0.013	0.006	0.100	PASS
39	0.017	0.007	0.100	PASS
40	0.012	0.005	0.100	PASS



Test setup

2.4.8 Test Location

This test was carried out in Harmonic & Flicker test room 118.

2.5 Flicker

2.5.1 Specification Reference

EN 61000-3-3:2013/A2:2021, Clause 5

2.5.2 Equipment Under Test

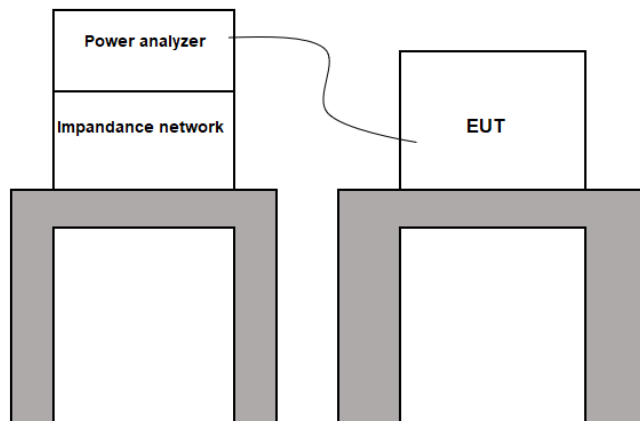
OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.5.3 Date of Test

12/19/2025, 12/23/2025, 01/29/2026

2.5.4 Test Method

For equipment not mentioned in annex A of EN 61000-3-3, controls or automatic programs should be set to produce the most unfavourable sequence of voltage change, using only those combinations of controls and programmes which are mentioned by the manufacturer in the instruction manual, or are otherwise likely to be used



2.5.5 Environmental Conditions

Ambient Temperature	10-35 °C
Relative Humidity	30-60 %
Atmospheric Pressure	1010-1040 mbar

2.5.6 Specification Limits

The value of P_{st} shall not be greater than 1.0

The value of P_{lt} shall not be greater than 0.65

T_{max} , the accumulated time value of $d(t)$ with a deviation exceeding 3.3% during a single voltage change at the EUT terminals, shall not exceed 500ms

The maximum relative steady-state voltage change, dc , shall not exceed 3.3%

The maximum relative voltage change d_{max} , shall not exceed

- a) 4% without additional conditions
- b) 6% for equipment which is:
 - Switched manually, or
 - Switched automatically more frequently than twice per day, and also has either a delayed start, or manual restart, after a power supply interruption
- c) 7% for equipment which is:
 - Attended whilst in use, or
 - Switched on automatically, or is intended to be switched on manually, no more than twice per day, and also has either a delayed restart or manual restart, after a power supply interruption

2.5.7 Test Results

Results for Configuration and Mode: AC Powered/switch on/off.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016PP26RN2 Power on, cooling, high fan speed, min temperature, timer on
IEC 61000-3-3 (Edition 3)

Test Standard : Limitation of voltage changes, voltage fluctuations and flicker
in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase
and not subject to conditional connection

Tester : Peiqiang WANG

Test Date : 12/19/2025 1:42:35 PM

Climatic Conditions

Temperature : 28 °C Pressure : 103 kPa Humidity : 58 %

Standard Group: Industry

Standard Name: IEC 61000-3-3 (Edition 3.2)
Limitation of voltage changes, voltage fluctuations and flicker
in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase
and not subject to conditional connection

Test Condition: Air conditioners, dehumidifiers, heat pumps, and commercial refrigerating equipment

Analysis Status: **PASS**

Flicker Measurements Settings

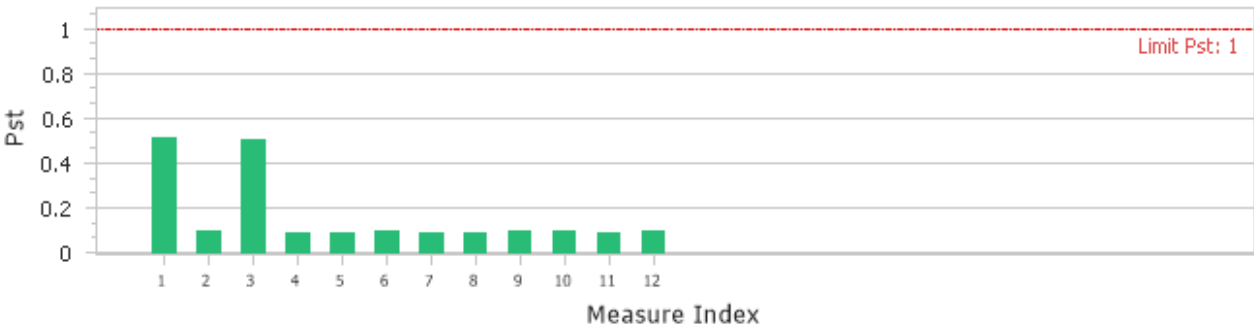
Main Line:	230V, 50Hz
Flicker Meter:	230V / 50Hz
Flicker Impedance:	Zref
Observation Time:	12 × 10 min
Measurements:	12

Flicker Measurements

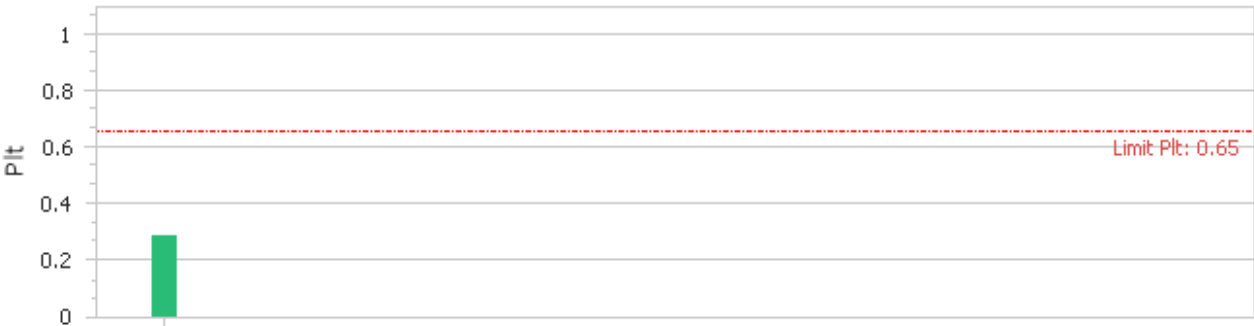
	P_{It}	Max P_{st}	Max d_c	Max d_{max}	Max T_{max}
Line 1:	0.287	0.521	0.559	3.392	0.01
Limits:	0.65	1	3.3	4	0.5
Results:	PASS	PASS	PASS	PASS	PASS



Short-term Flicker Severity (Pst) (Line 1)



Long-term Flicker Severity (Plt) (Line 1)



Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016JA26RN2 Power on, cooling, high fan speed, min temperature, timer on
IEC 61000-3-3 (Edition 3)

Test Standard : Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection

Tester : Peiqiang WANG

Test Date : 12/23/2025 1:06:56 PM

Climatic Conditions

Temperature : 28 °C Pressure : 103 kPa Humidity : 57 %

Standard Group: Industry

Standard Name: IEC 61000-3-3 (Edition 3)
Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection

Test Condition: Air conditioners, dehumidifiers, heat pumps, and commercial refrigerating equipment

Analysis Status: **PASS**

Flicker Measurements Settings

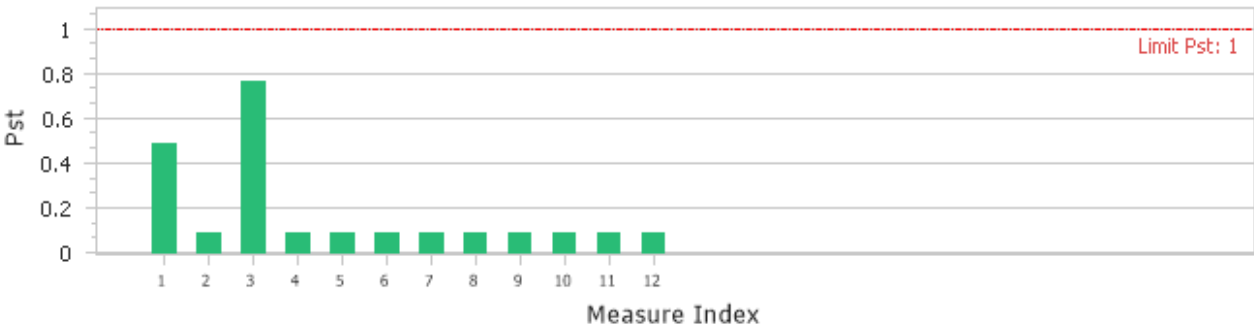
Main Line:	230V, 50Hz
Flicker Meter:	230V / 50Hz
Flicker Impedance:	Zref
Observation Time:	12 × 10 min
Measurements:	12

Flicker Measurements

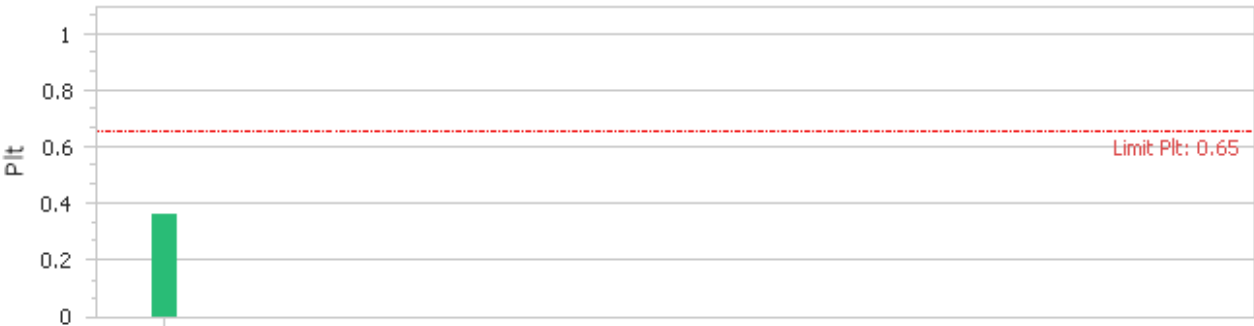
	P_{It}	Max P_{st}	Max d_c	Max d_{max}	Max T_{max}
Line 1:	0.366	0.771	2.896	3.103	0
Limits:	0.65	1	3.3	4	0.5
Results:	PASS	PASS	PASS	PASS	PASS



Short-term Flicker Severity (Pst) (Line 1)



Long-term Flicker Severity (Plt) (Line 1)



Test Report

Report Number : 878882357001-03

Test Object : Portable air conditioner OL-A016JC26RN2/S Power on, cooling, high fan speed, min temperature, timer on, swing
IEC 61000-3-3 (Edition 3)

Test Standard : Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection

Tester : Peiqiang WANG

Test Date : 1/29/2026 8:15:55 PM

Climatic Conditions

Temperature : 28 °C Pressure : 103 kPa Humidity : 58 %

Standard Group: Industry

Standard Name: IEC 61000-3-3 (Edition 3.2)
Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection

Test Condition: Air conditioners, dehumidifiers, heat pumps, and commercial refrigerating equipment

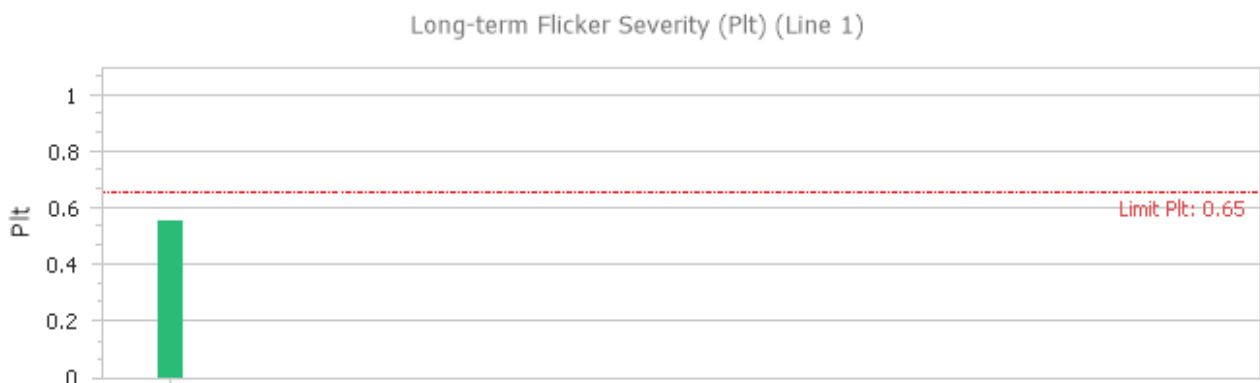
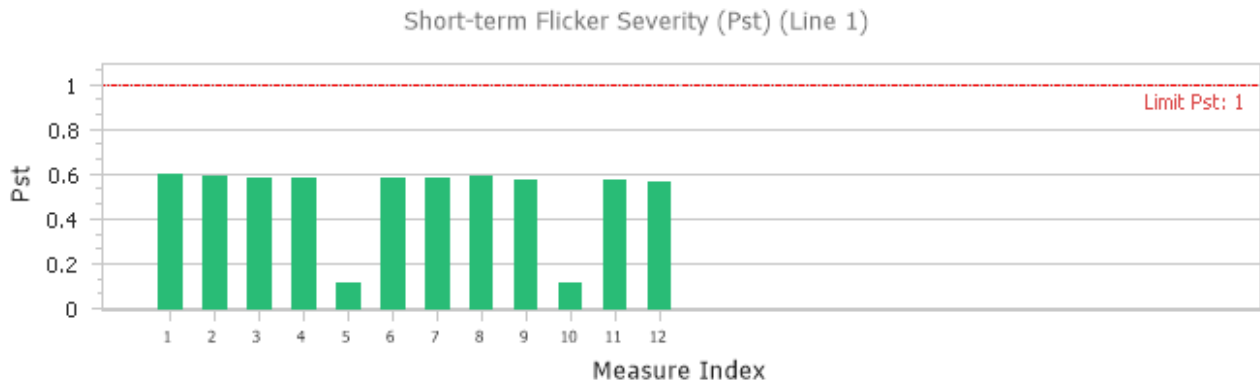
Analysis Status: **PASS**

Flicker Measurements Settings

Main Line:	230V, 50Hz
Flicker Meter:	230V / 50Hz
Flicker Impedance:	Zref
Observation Time:	12 × 10 min
Measurements:	12

Flicker Measurements

	P _{It}	Max P _{st}	Max d _c	Max d _{max}	Max T _{max}
Line 1:	0.554	0.606	0.509	3.921	0.1
Limits:	0.65	1	3.3	4	0.5
Results:	PASS	PASS	PASS	PASS	PASS



Test setup

2.5.8 Test Location

This test was carried out in Harmonic & Flicker test room 118.

2.6 Electrostatic discharge immunity test

2.6.1 Specification Reference

EN IEC 55014-2:2021, Clause 5.1

2.6.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.6.3 Date of Test

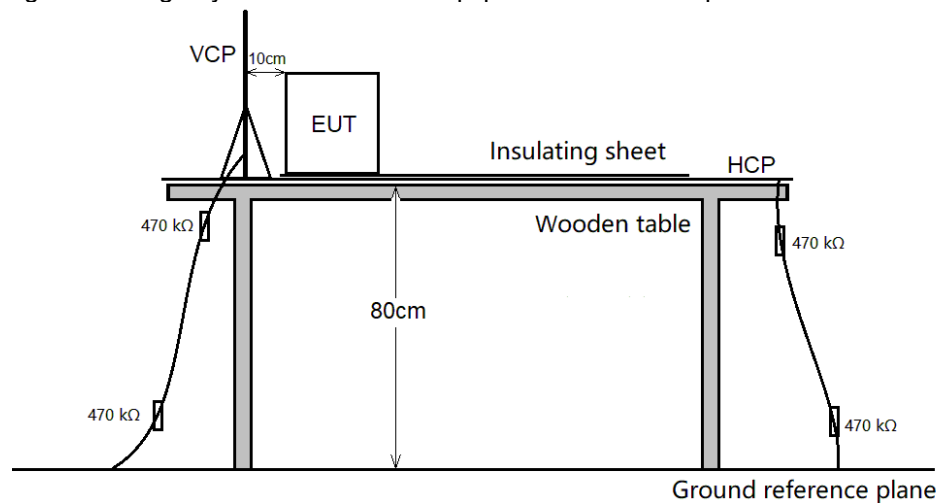
12/31/2025, 02/02/2026, 02/02/2026

2.6.4 Test Method

The equipment under test including associated cabling was configured on but insulated from, using a 0.5mm isolator, a horizontal coupling plane fitted to the top of a 0.8m non-conductive table for table-top equipment; and on a 0.1m insulated support for floor standing equipment; above a ground reference plane all within a test laboratory.

Using the air discharge method for non-metallic parts, contact discharge method for metallic parts with both vertical and horizontal couple plane discharge methods for the sides of the equipment under test, the required electrostatic discharge voltage levels in both voltage polarities were applied at the detailed pulse repartition rate.

During this testing any anomalies in the equipment under tests performance was recorded.



2.6.5 Environmental Conditions

Ambient Temperature 22.3 °C
Relative Humidity 52.1 %
Atmospheric Pressure 1025.0 mbar

2.6.6 Specification Limits

Required Test Levels				Performance Criteria
Discharge type	Discharge Level (kV)		Number of discharges per location (each polarity)	
	Positive	Negative		
Air – Direct	8	8	<10>	B
Contact – Direct	4	4	<10>	B
Contact – Indirect	4	4	<10>	B

2.6.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

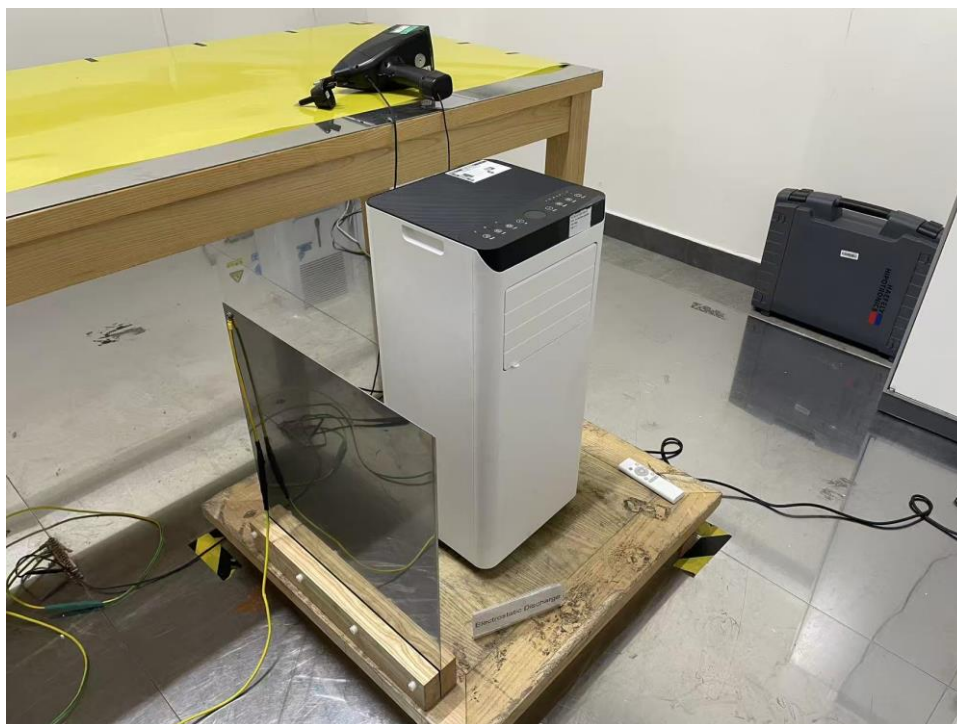
Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

ID	Test Point	Discharge	Results									
			2kV		4kV		6kV		8kV		15kV	
			+	-	+	-	+	-	+	-	+	-
A	Vertical coupling plane	Contact			✓	✓						
B	Screw	Contact			✓	✓			✓	✓		
C	Plastic enclosure	Air							✓	✓		
D	Gap	Air							✓	✓		

Note:

✓	The EUTs performance was not impacted when the ESD pulse was applied.
✓*	No discharge occurred at this point when the ESD pulse was applied.
Ox	Observation number A, B, ...etc.



Test Setup

2.6.8 Test Location

This test was carried out in ESD room 117.

2.7 Electrical fast transient /burst immunity test

2.7.1 Specification Reference

EN IEC 55014-2:2021, Clause 5.2

2.7.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.7.3 Date of Test

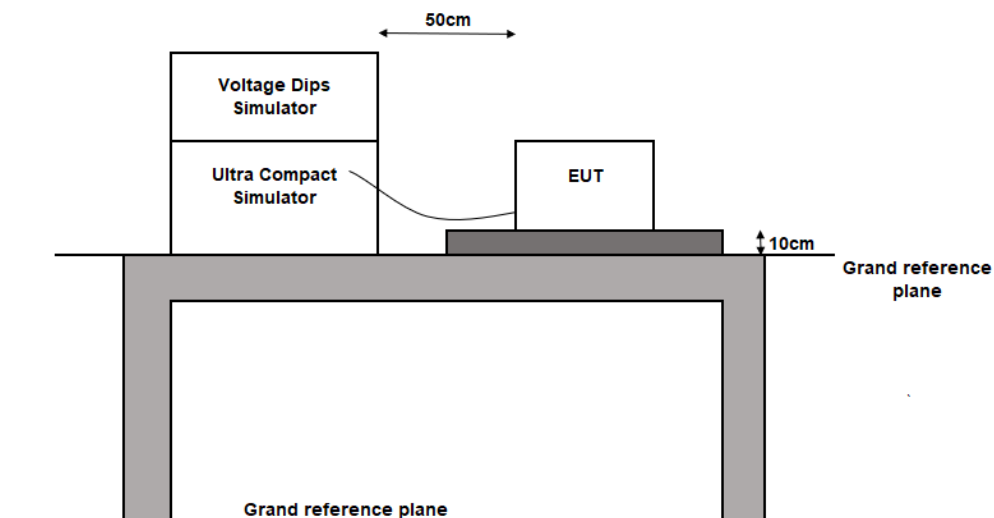
12/31/2025, 02/02/2026

2.7.4 Test Method

The equipment under test including associated cabling was configured on but insulated from, using a 0.1 m isolator, a horizontal coupling plane fitted to the top of a 0.8 m non-conductive table for table-top equipment; and on a 0.1 m insulated support for floor standing equipment; above a ground reference plane all within a test laboratory.

Using a CDN for power ports, capacitive coupling clamp for signal and control ports and a 33Nf coupling capacitor for earth ports, the required fast transient burst voltage levels in both voltage polarities were applied at the detailed pulse repartition rate and duration of test.

During this testing any anomalies in the equipment under tests performance was recorded.



2.7.5 Environmental Conditions

Ambient Temperature	20-25 °C
Relative Humidity	40-60 %
Atmospheric Pressure	1010-1040 mbar

2.7.6 Specification Limits

Required Test Levels Input and output a.c. power ports					Performance Criteria
Line Under Test	Level (kV)	Repetition Rate (kHz)	Test Duration	Coupling Method	
AC power port	± 1	5 kHz	2 min per polarity	CDN	B
For extra low voltage a.c. ports and output a.c. ports, this testing is only applicable to ports interfacing with cables whose total length may exceed 3m according to the manufacturer's functional specification.					

2.7.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Results for Fast Transient Burst Immunity					
Line under test	Test Level (kV)	Repetition Rate	Test Duration	Coupling Method	Result
Power line	± 1.0	5 kHz	2 min	CDN	Pass PC A



Test Setup

2.7.8 Test Location

This test was carried out in shielded room 121.

2.8 Immunity to conducted disturbances, induced by radio-frequency fields

2.8.1 Specification Reference

EN IEC 55014-2:2021, Clause 5.3

2.8.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.8.3 Date of Test

12/31/2025, 02/02/2026

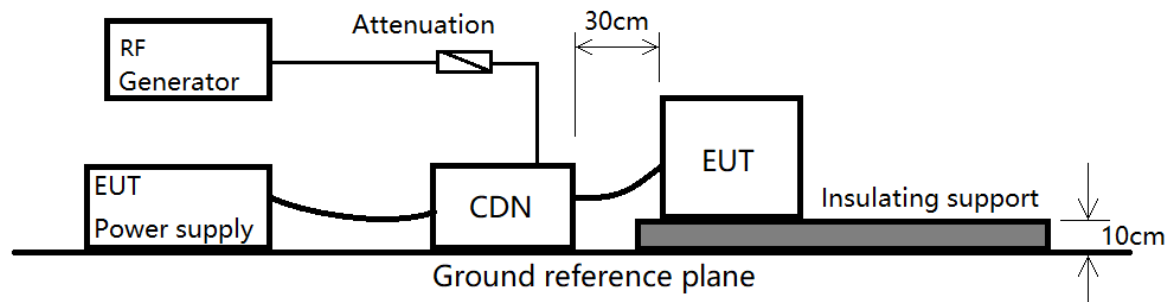
2.8.4 Test Method

The equipment under test was placed on an insulating support 0,1 m above the reference ground plane.

All associated cabling was configured, on but insulated from, using a 50 mm isolator, the same horizontal coupling plane as the equipment under test.

Using CDNs, EM Clamps or current clamps as appropriate, the power ports and applicable signal and control ports were subjected to the required, pre calibrated RF injected signal strength, modulated as described, swept over the frequency range of test.

During this testing any anomalies in the equipment under tests performance was recorded.



2.8.5 Environmental Conditions

Ambient Temperature 20-25 °C
Relative Humidity 40-60 %
Atmospheric Pressure 1010-1040 mbar

2.8.6 Specification Limits

Required Test Levels Input and output a.c. power ports						Performance Criteria
Line Under Test	Frequency Range (MHz)	Level (V)	Modulation	Step Size (%)	Dwell (s)	
AC power port	0.15 to 230	3	AM (80 %, 1 kHz, sine wave)	1	3	A
For extra low voltage a.c. ports and output a.c. ports, this testing is only applicable to ports interfacing with cables whose total length may exceed 3m according to the manufacturer's functional specification.						

2.8.7 Test Results

Results for Configuration and Mode:

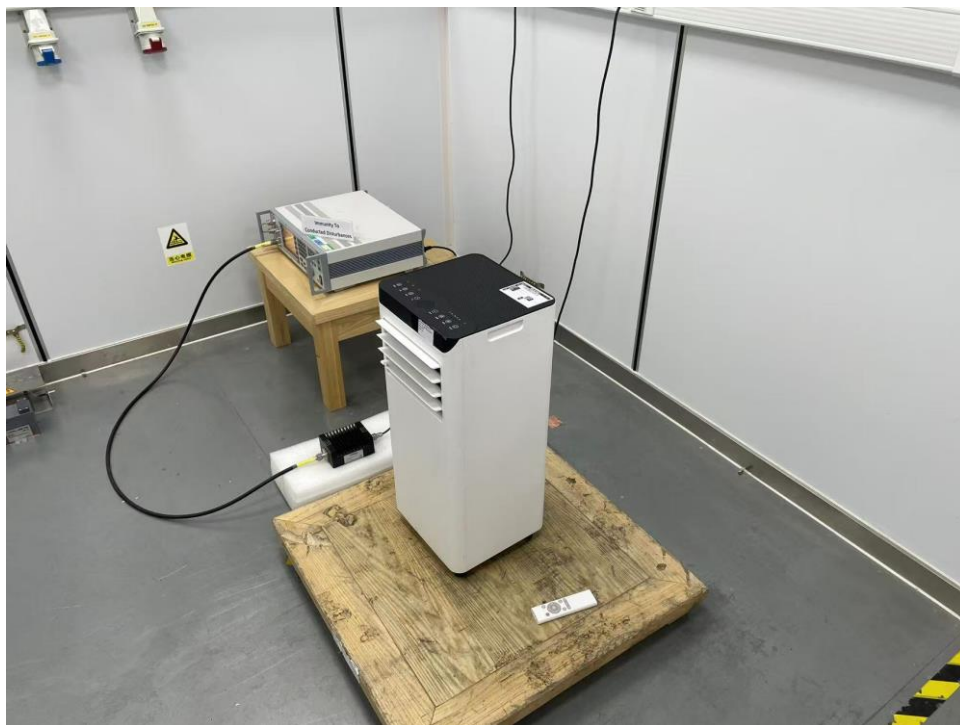
OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Results for Injected current						
Line under test	Test Level	Step	Dwell Time	Coupling Method	Modulation	Result
Power line	3V	1%	3S	CDN	1KHz 80%	Pass PC A



Test Setup

2.8.8 Test Location

This test was carried out in in shielded room 121.

2.9 Radiated, radio-frequency, electromagnetic field immunity test

2.9.1 Specification Reference

EN IEC 55014-2:2021, Clause 5.5

2.9.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.9.3 Date of Test

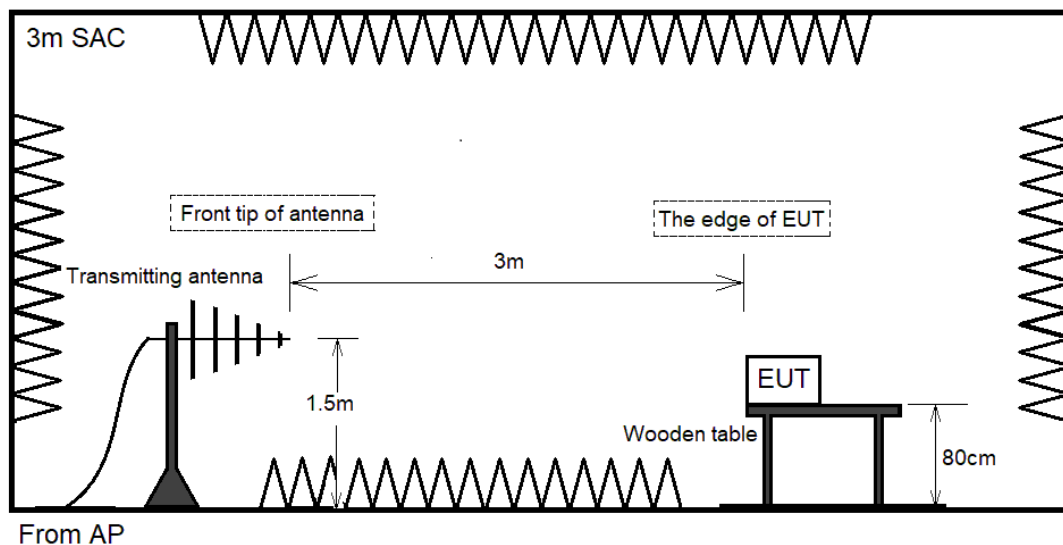
12/31/2025, 02/02/2026

2.9.4 Test Method

The equipment under test including associated cabling was configured, on a 0.8 m non-conductive table for table-top equipment and on a 0.1 m insulated support for floor standing equipment; with a pre-calibrated semi anechoic chamber.

All four side of the equipment under test were subjected to the required RF field strength, modulated as described, swept over the frequency range of test with the antenna positioned in both horizontal and vertical polarizations.

During this testing any anomalies in the equipment under tests performance was recorded.



2.9.5 Environmental Conditions

Ambient Temperature 20-25 °C
 Relative Humidity 40-60 %
 Atmospheric Pressure 1010-1040 mbar

2.9.6 Specification Limits

Frequency Range (MHz)	Required Test Levels				Performance Criteria
	Level (V/m)	Modulation	Step Size (%)	Dwell (s)	
80 to 1000	3	AM (80 %, 1 kHz, sine wave)	1	3	A
Note 1. EUT powered at one of the Nominal input voltages and frequencies					

2.9.7 Test Results

Results for Configuration and Mode:

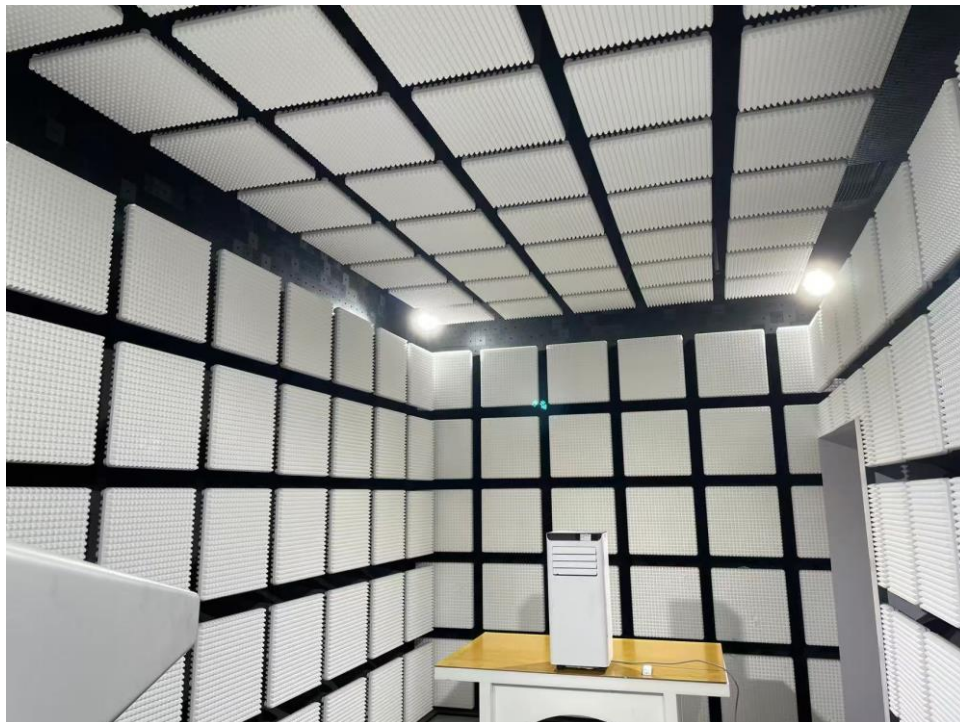
OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Results for RF Electromagnetic Field 80 - 1000 MHz				
Side of the equipment under test	Antenna polarization	Test Level	Dwell Time	Result
Front	horizontal	3 V/m	3 s	Pass PC A
Front	vertical	3 V/m	3 s	Pass PC A
Left	horizontal	3 V/m	3 s	Pass PC A
Left	vertical	3 V/m	3 s	Pass PC A
Rear	horizontal	3 V/m	3 s	Pass PC A
Rear	vertical	3 V/m	3 s	Pass PC A
Right	horizontal	3 V/m	3 s	Pass PC A
Right	vertical	3 V/m	3 s	Pass PC A



Test Setup

2.9.8 Test Location

This test was carried out in 3m anechoic chamber.

2.10 Surge immunity test

2.10.1 Specification Reference

EN IEC 55014-2:2021, Clause 5.6

2.10.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

2.10.3 Date of Test

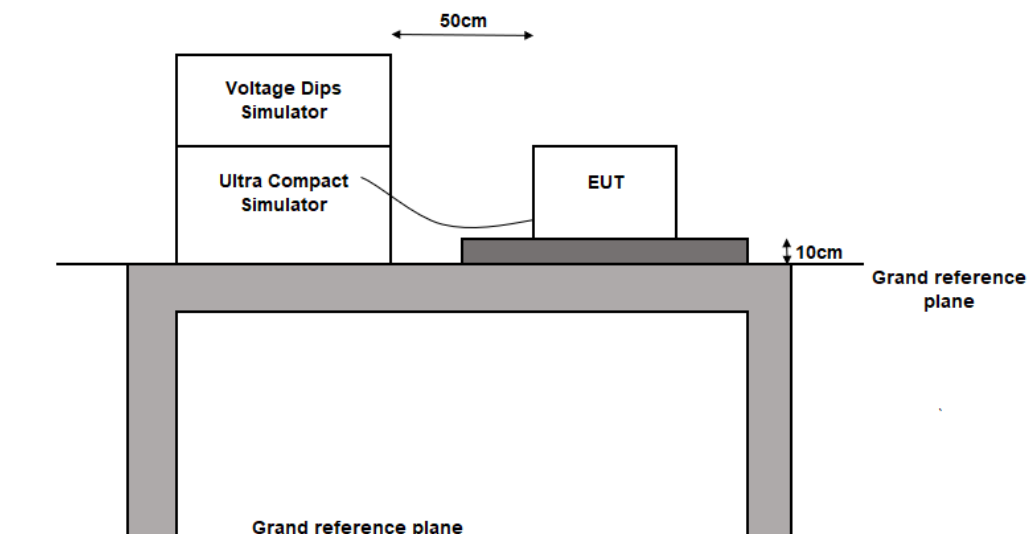
12/31/2025, 02/02/2026

2.10.4 Test Method

The equipment under test including associated cabling was configured, on a 0.8 m non-conductive table for table-top equipment and on a 0.1 m insulated support for floor standing equipment above a ground reference plane all within a test laboratory.

Using CDNs for power ports and appropriate coupling methods for applicable signal and control ports, the required number of surges was applied for each surge voltage level using both positive and negative surge voltage polarities. Surges were applied at the power line frequency phase angles and repartition rates detailed.

During this testing any anomalies in the equipment under tests performance was recorded.



2.10.5 Environmental Conditions

Ambient Temperature 20-25 °C
 Relative Humidity 40-60 %
 Atmospheric Pressure 1010-1040 mbar

2.10.6 Specification Limits

Characteristics	Test Levels	Performance Criteria
Wave- shape data Test levels line to line with 2Ω impedance line to earth with 12Ω impedance	1.2/50 μs ± 1.0 kV ±2.0 kV	B
Note In addition to the specified test level, all lower levels as detailed in IEC 61000-4-5 should also be satisfied.		

2.10.7 Test Results

Results for Configuration and Mode:

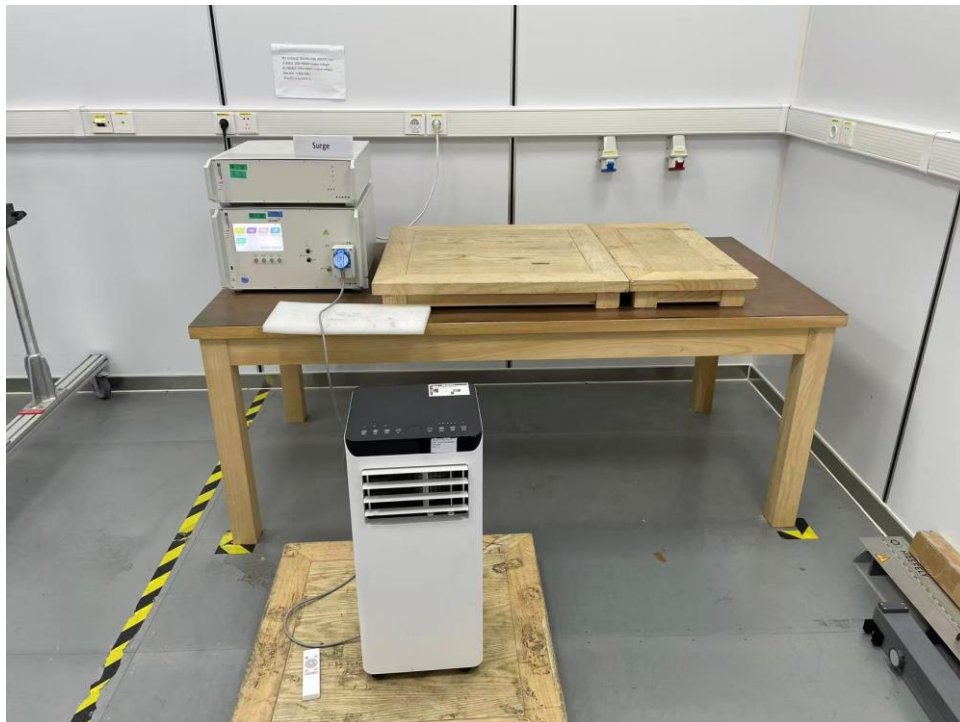
OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Results for Surge Immunity (Power Ports)							
Line Name	Coupling	Level	Polarity	Phase Angle	No of Pulses	Repetition Rate	Result
Power line	Live to Neutral	-1.0kV	NEGATIVE	270 deg	5	60 sec	Pass PC A
Power line	Live to Neutral	+1.0kV	POSITIVE	90 deg	5	60 sec	Pass PC A
power line	Live to Earth	-2.0kV	NEGATIVE	270 deg	5	60 sec	Pass PC A
power line	Live to Earth	+2.0kV	POSITIVE	90 deg	5	60 sec	Pass PC A
power line	Neutral to Earth	-2.0kV	NEGATIVE	270 deg	5	60 sec	Pass PC A
power line	Neutral to Earth	+2.0kV	POSITIVE	90 deg	5	60 sec	Pass PC A



Test Setup

2.10.8 Test Location

This test was carried out in in shielded room 121.

2.11 Voltage dips, short interruptions and voltage variations immunity tests

2.11.1 Specification Reference

EN 55014-2:2015, Clause 5.7

2.11.2 Equipment Under Test

OL-A016PP26RN2, OL-A016JA26RN2, OL-A016JC26RN2/S

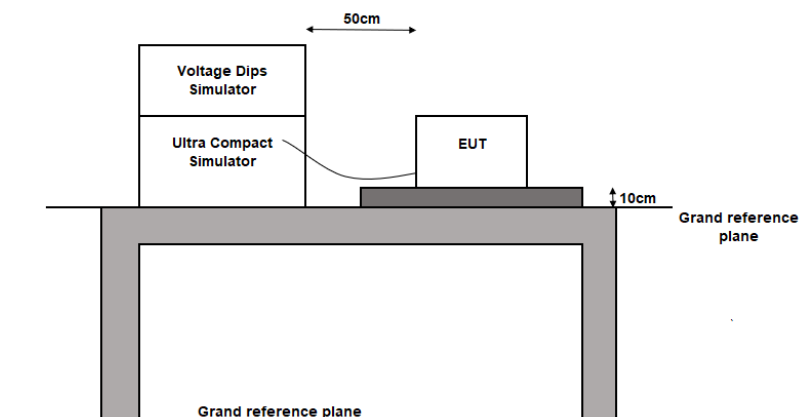
2.11.3 Date of Test

12/31/2025, 02/02/2026

2.11.4 Test Method

The equipment under test including associated cabling was configured, on a 0.8 m non-conductive table for table-top equipment and on a 0.1 m insulated support for floor standing equipment above a ground reference plane all within a test laboratory.

Using a programmable power supply the equipment under test was subjected to the detailed supply voltage dips and interruptions. The required supply phase synchronization and test repetition rate, detailed, was controlled by the programmable power supply. During this testing any anomalies in the equipment under tests performance was recorded.



2.11.5 Environmental Conditions

Ambient Temperature 20-25 °C
Relative Humidity 40-60 %
Atmospheric Pressure 1010-1040 mbar

2.11.6 Specification Limits

Voltage dips				
Voltage Dip in % Ut	Test level in % Ut	Duration		Performance Criteria
		50Hz	60Hz	
100	0	½ cycle	½ cycle	C
60	40	10 cycles	12 cycles	C
30	70	25 cycles	30 cycles	C
Ut is the rated voltage of the Equipment Under Test				

2.11.7 Test Results

Results for Configuration and Mode:

OL-A016PP26RN2, OL-A016JA26RN2: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on;

OL-A016JC26RN2/S: AC Powered/Power on, all function, high & low fan speed, mid temperature, timer on, swing.

Performance assessment of the EUT made during this test: Pass.

Detailed results are shown below.

Test Results for Voltage Dip and Short Interruption					
Line under test	Vnom		Test Level	Duration	Result
Power line	230 Vac	50 Hz	0% of Vnom	½ cycle	Pass PC B
Power line	230 Vac	50 Hz	40% of Vnom	10 cycles (50Hz)	Pass PC B
Power line	230 Vac	50 Hz	70% of Vnom	25 cycles (50Hz)	Pass PC B
B means during the test, the EUT would stop working, once remove the interference, the EUT can restore its original mode by itself.					



Test Setup

2.11.8 Test Location

This test was carried out in shielded room 121.

3 Test Equipment Information

3.1 General Test Equipment Used

Instrument	Manufacturer	Type No	TE No	Calibration Date	Calibration Due
Conducted Emission (Discontinuous Disturbance)					
EMI test receiver	R & S	ESR3	NIB1910011-YQ-EMC	2025.4.13	2026.4.12
2-Line V-network	R & S	ENV216	NIB1910012-YQ-EMC	2025.4.13	2026.4.12
Radiated Disturbance (30MHz to 1000MHz)					
EMI Test Receiver	R&S	ESR7	NIB1910038-YQ-EMC	2026.1.15	2027.1.14
TRILOG Super Broadband Test Antenna	SCHWARZBEC K	VULB 9168	NIB1910045-YQ-EMC	2025.3.13	2026.3.12
3 meter semi-anechoic chamber	TDK	3m	NIB2101059-YQ-EMC	2025.4.13	2030.4.12
Harmonic current emission and Flicker					
Harmonic-flicker test system	EM test	DPA 500N	NIB1911001-YQ-EMC	2025.4.13	2026.4.12
AC/DC power source	EM test	NetWave 7-400	NIB1911002-YQ-EMC	2025.4.13	2026.4.12
Electrostatic discharge immunity test					
ESD Simulator	HAEFELY	ONYX 16	NIB1911018-YQ-EMC	2025.10.24	2026.10.23
Temperature and Humidity Data Logger	Testo	175H1	NIB2005022-YQ-EMC	2025.4.18	2026.4.17
Barometer	VAREN	132B	NIB2005029-YQ-EMC	2025.4.21	2026.4.20
Vertical Coupling Plane	TÜV Product Service	---	---	---	---
Radiated, radio-frequency, electromagnetic field immunity test					
Signal Generator	R&S	SMB100B	NIB1910039-YQ-EMC	2026.1.15	2027.1.14
Power Meter	R&S	NRX	NIB1910040-YQ-EMC	2026.1.15	2027.1.14
Average Power Sensor	R&S	NRP6A	NIB1910043-YQ-EMC	2026.1.15	2027.1.14
Average Power Sensor	R&S	NRP6A	NIB1910044-YQ-EMC	2026.1.15	2027.1.14
Broadband Amplifier	R&S	BBA150-BC250	NIB1910037-YQ-EMC	2026.1.19	2027.1.18
Broadband Antennas	SCHWARZBECK	STLP9129	NIB1910047-SB-EMC	---	---
Open switch and control platform 2HU	R&S	OSP220	NIB1910042-SB-EMC	---	---

Instrument	Manufacturer	Type No	TE No	Calibration Date	Calibration Due
Electrical fast transient/burst immunity test					
Ultra compact simulator	HAEFELY	AXOS8	NIB1911019-YQ-EMC	2025.4.13	2026.4.12
Immunity to conducted disturbances, induced by radio-frequency field					
Signal generator	TESEQ	NSG4070C-80	NIB1911003-YQ-EMC	2025.4.13	2026.4.12
6dB attenuator	TESEQ	ATN 6060	NIB1911007-SB-EMC	---	---
Coupling Decoupling Network	TESEQ	CDN M016	NIB1911006-YQ-EMC	2025.4.13	2026.4.12
Surge immunity test					
Ultra compact simulator	HAEFELY	AXOS8	NIB1911019-YQ-EMC	2025.4.13	2026.4.12
Voltage dips, short interruptions and voltage variations immunity test					
Ultra compact simulator	HAEFELY	AXOS8	NIB1911019-YQ-EMC	2025.4.13	2026.4.12
Voltage Dips Simulator	HAEFELY	DIP116	NIB1911020-YQ-EMC	2025.4.13	2026.4.12

3.2 EMC Testing Software

Software	Version	Testing Items
EMC 32	V10.60.20	CE, RE, RS
Click Rate Analyzer	V2.5.1	Click
NET.CONTROL	V3.0.3	Harmonic and Flicker

4 Measurement Uncertainty

For a 95% confidence level, the measurement uncertainties for defined systems are:

Test Name	Measurement Uncertainty
Conducted Disturbance	150kHz to 30 MHz, 2.66dB (AMN)
Radiated Disturbance	30MHz to 1GHz, 4.18dB (Horizontal) 5.45dB (Vertical)

Measurement Uncertainty Decision Rule

Determination of conformity with the specification limits is based on the decision rule according to IEC Guide 115: 2023, clause 4.3.3.

5 Photographs

OL-A016PP26RN2





OL-A016JA26RN2





OL-A016JC26RN2/S





Remote control (OL-YRQ-002)



Remote control (OL-YRQ-004)

